QMX+ Panadapter — User Guide (v1.3.0)

By Steffen Lav (OZ1LAV). Generated from the project README — for the live page, issues, and releases see https://github.com/SteffenLav/qmx-panadapter

---

The full documentation also lives online at **tab5.lav.dk** — the user guide, quick-start, and reference as searchable, cross-linked web pages, kept in sync with every release.

Features

FT8 digital mode — Full-featured FT8 station (15-second slots, 79 symbols @ 160 ms cadence): continuous on-device RX decoding, TX to reply or call CQ, auto-QSO workflows, ADIF logging — all in the same panadapter view. **FT4** (7.5-second slots, 105 symbols @ 48 ms cadence) is fully supported too, re-enabled in v0.21.0 once the FT8/FT4 render gate freed the processor headroom its faster cadence needs.

Real-time panadapter — Spectrum and waterfall with tap-to-tune, pinch-zoom, and touch-drag one-finger navigation. 30 Hz refresh, 12 kHz IF offset compensation, flat-spectrum mode, adaptive waterfall floor, FFT window selection, and S-meter. Screenshot and spectrum export to web browser.

QMX CAT control — Live frequency readout, mode switching (USB/LSB/CW/DiGi), SSB filter bandwidth control, passband indicator, and TX power/SWR readout. Round-trip CAT latency <50 ms. Band presets and per-band frequency recall.

ADIF logging — Every QSO logs to onboard storage with QSO timestamp, callsign, frequency, mode, signal report, grid, and distance. Export to web UI or QRZ Logbook / eQSL for cloud backup.

microSD station backup — Insert a microSD card (a plain FAT32 32 GB card is ideal) and the Tab5 automatically mirrors your whole station to `/qmx-panadapter/` on the card: the ADIF QSO log, a full config export (settings + memory channels), your LoTW signing certificate + key, the diagnostic log, and a self-describing `README.txt`. No setup required — a green **SD** dot in the bottom bar lights when a card is mounted. A genuine grab-and-go backup for PC-free POTA/SOTA. *(The card holds credentials — WiFi password, QRZ/eQSL logins, LoTW private key — so keep it physically secure.)*

Web UI — Browser panadapter, remote control (tune, mode, bandwidth), QSO log viewer, config export/import, microSD file browser, and diagnostic log download — all without leaving the radio room.

USB mouse — Plug a mouse into the USB-A port and a cursor appears; clicks drive every menu, button and drawer. *One hardware limitation: the mouse and the QMX can't share the single USB host port (a hub doesn't help — the ESP32-P4's USB stack lacks the Transaction Translator a high-speed hub needs for them), so the mouse is for setup, log review, and reading the manual with the radio unplugged.*

Offline capable — WiFi optional. For POTA/SOTA use cases: Tab5 RTC keeps time across power-off, FT8/FT4 timing stays locked via internal oscillator + periodic QMX sync (no GPS required).

Contents

Chapters

Appendices

1. Quick Guide

Step 0 — Check your QMX firmware first

Power the QMX on by itself and read the firmware version off its own display at boot. You need **1.03.002 or newer**. If yours is older, update the QMX *before* connecting the Tab5 — everything that follows depends on it. This takes 10 seconds to verify and saves hours of debugging. Both **1.03.002** and the **1.04.002 beta** work with the panadapter; on 1.04.002 the Tab5 additionally offers **AM mode** and an **Antenna Tune** button (SWR tune with a live power/SWR readout) — both stay hidden on 1.03.002, so there's no downside either way.

Step 1 — Flash the Tab5 firmware

Use the one-click flasher in `tools/QMX-Panadapter flasher/` (also attached to each release):

1. Plug the Tab5 into your computer with a **USB-C data cable** — charge-only cables will not work.
2. Run the flasher:
   - **Windows** — double-click `flash.bat` (downloads esptool + firmware automatically; nothing to install)
   - **macOS** — double-click `flash.command` (needs esptool once — `brew install esptool` recommended; `pip3 install esptool` often fails on recent macOS with an "externally-managed-environment" error). If macOS blocks the script, right-click → **Open**, or run `bash flash.command` in Terminal (no `chmod` needed).
   - **Linux** — `bash flash.command` (needs `pip3 install esptool`)
3. The flasher asks **normal or clean** (see below) — just press **Enter** for a normal flash.
4. Wait for `SUCCESS`. The Tab5 restarts on the new firmware.

The flasher downloads the latest release from GitHub automatically. **No internet?** Put a `qmx_panadapter_merged_*.bin` from the releases page next to the flasher and it uses that instead.

Normal vs clean flash. Just before flashing, the flasher asks:

> *Type E for a CLEAN/ERASE flash, or just Enter for a normal flash*

- **Normal (press Enter)** — updates the firmware and **keeps** all your saved settings (WiFi, callsign, grid, memory channels, ADIF log). This is what you want almost every time.

- **Clean (type E)** — wipes the whole chip first, so **every saved setting is permanently erased**: WiFi name *and* password, callsign/grid, all memory channels, and the logged QSOs. Use it only if something is stuck or corrupted (e.g. WiFi refuses to turn on no matter what). Back up first with **Config ↓** (below) so you can restore in seconds afterwards.

Building from source? See Build from source.

Step 2 — Connect the cables

You need **two** USB connections, and the cable to the QMX is the one people get wrong:

Connection	Port on Tab5	Cable	Carries
Tab5 → QMX	**USB-A** (host)	**USB-A → USB-C, full data cable**	I/Q audio (UAC) + CAT (CDC-ACM)
Tab5 → power	**USB-C**	Any USB-C power cable (5 V / 2 A+)	Power

> **The #1 failure is a charge-only cable.** Many USB-C cables — especially thin ones bundled with phones and chargers — carry power only, no data lines. If you use one between the Tab5's USB-A port and the QMX, the Tab5 powers the QMX but sees no audio and no CAT: the spectrum stays flat and the top bar shows `Band: ---`. Use a cable you know does data (one that works for a USB stick or phone file transfer). When in doubt, swap the cable first.

Power-on order matters. Turn the **Tab5 on first** and let it finish loading, then turn the **QMX on**. Within a few seconds the top bar should populate Band / Mode / BW and the spectrum should come alive.

Once flashed you can power the Tab5 from any 5 V/2 A USB-C source or the internal battery — the laptop is only needed for flashing. For diagnostics the Tab5 also outputs a serial log over the USB-C data connection (useful if WiFi is not available), but for most users the built-in **diagnostic log** — always on, nothing to enable — is the easier path — see Step 7.

Step 3 — Find your way around

The whole app is driven by **one-finger swipes from the screen edges** and **taps on the top bar**. The settings drawer is always one swipe away — no hunting for hidden menus.

Gesture	From	Does
Swipe →	Left edge	Toggle Panadapter ↔ FT8 screen
Swipe ←	Right edge	Open settings drawer
Swipe ↑	Bottom edge	Open memory channel picker
Tap or swipe ↓	Any top-bar item	Open that item's selector

Slim "breathing" grip handles on each edge show where to swipe — faint enough not to clutter the display, visible if you look for them.

The top bar reads **Band · Mode · BW · Freq · Signal · Zoom** left to right. Tap any item to open a selector — **Freq** opens the frequency keypad; **Mode** switches USB/LSB/CW/DiGi; **BW** selects SSB filter width (USB/LSB only); **Band** jumps to a configured band; **Zoom** selects a zoom preset.

See the full gesture table in the Reference section.

Step 3b — Choose FT8 or FT4 mode

FT8 is available on all HF bands; FT4 is available on select bands with published frequencies (20 m, 30 m, 40 m, etc.). To switch between them:

1. **In Panadapter mode:** Tap the **Mode** selector in the top bar, then choose **DiGi** (the digital mode group).
2. **In FT8 screen:** Tap the **Preset** frequency button to open the band/mode picker — you'll see **two columns**: FT8 (gold header) on the left, FT4 (cyan header) on the right. Tap any frequency in the FT4 column to switch modes.

The decode list, TX controls, and slot timing (countdown bar, which-parity-next) all adapt automatically to the active mode. Switching modes clears stale decodes from the previous mode so you don't work old QSOs.

When to use each mode:

- **FT8** — Most populated band in SOTA/POTA; longer decode window gives more time to prepare replies on slow links or weak signals.

- **FT4** — Faster QSO cycle for busy bands (Field Day, major contests, 20 m pile-ups) and operators who prefer shorter waits between exchanges.

Step 4 — Fill in your settings

On first boot the Tab5 opens a setup prompt automatically — it will ask for your callsign, grid square, and WiFi credentials before you reach the main screen. You can skip any field and fill it in later via the settings drawer.

To open the settings drawer at any time: swipe in from the right edge, or tap the right grip handle.

- **WiFi** — enter your SSID and password. Recommended for everyone: you get accurate UTC time (needed for FT8 slot timing), the browser web UI, and remote diagnostics. The **WiFi initiated** checkbox below the WiFi section is an on/off master switch — uncheck it for POTA or field use with no network.
- **Identity** (callsign + grid square) — *only required for FT8/FT4 transmit*. Enter your callsign and Maidenhead grid (4 or 6 characters, e.g. `JO45` or `JO45ab`). The grid also drives the distance and bearing columns in the FT8 decode list.

Everything else — dB range, smoothing, colour map, brightness, IQ balance — has sane defaults. Adjust if you want to, but you don't need to on first use.

Step 5 — Using the panadapter

Switch the QMX to any band and watch it come alive. This is the foundation of the device regardless of which modes you operate.

Tap or drag to tune. Touch anywhere on the spectrum or waterfall to place the cyan cursor. Drag and it snaps to a mode-aware grid — 10 Hz steps in CW, 250 Hz in SSB, 500 Hz for FT8/FT4 — so you can land precisely on a signal before lifting your finger. Lift, and the QMX retunes.

Swipe fast to pan instead. A quick horizontal swipe (rather than a held drag) slides the whole view left or right and retunes to wherever you let go — the fastest way to slide along a band. See Touch-to-tune for exactly how the two gestures are told apart.

Or move precisely with the band-plan slider. For a controlled move rather than a quick swipe, drag the framed "visible window" on the band-plan strip (along the bottom) to exactly where you want to be on the band — the spectrum, waterfall, and VFO all follow it, so you can place yourself precisely on a segment before you even tap a signal.

Zoom in on a crowded band. Pinch with two fingers to zoom up to ×24. On a CW band at ×8 or higher you can resolve individual signals a few hundred Hz apart, read the spacing between callers, and pick your target before you tune. The frequency axis scales with you, down to Hz precision at high zoom. Double-tap anywhere to reset to the full 48 kHz view.

Find your place on the band. The thin band-plan strip along the bottom of the screen (just above the status bar) colour-codes the CW/Digi/Phone segments around you — pick your IARU region (or leave it on Auto) in the settings drawer → **Band-plan region**. The framed window on the strip is a **slider**: drag it (or tap anywhere on the strip) to move precisely to any spot on the band, and the spectrum, waterfall and VFO all follow along. You can also grab that slider from anywhere along the bottom status bar and drag sideways — a taller target — while a vertical up-swipe there still opens memory channels.

Read your passband. Two grey vertical lines on the spectrum mark your current filter edges. The BW label in the top bar shows the active width; tap it to choose 2.5 / 2.7 / 2.9 / 3.2 kHz in USB or LSB. A coloured tint fills the passband so you can always see exactly what slice of the band you're receiving.

Watch the S-meter. The Signal field in the top bar is a live tick-scale bar (S1 through S9+20), driven by the actual signal under your VFO cursor.

Flat-spectrum mode. Toggle **Flat spectrum** in the settings drawer to switch from absolute dBm to dB-above-local-noise-floor. Noise collapses to a calm baseline; real signals — including weak CW tones in the mud — pop sharply above it. Recommended on noisy bands.

Save your spots. Swipe up from the bottom edge to open the memory channel picker — a 4×8 grid of 32 channels. It doesn't greet you with blank cells: a fresh device arrives with a handful of **example channels already filled in**, and each one is **colour-coded by mode** (CW green, DiGi teal, USB brick-red, LSB purple) so you can read the whole grid at a glance without opening a single label. The very first time you open it, a quick **guided animation** walks you through the gestures — watch a channel slide to a new slot, then drop into the wastebin and vanish — so you learn "drag to move, drag to delete" in about ten seconds, once, and never again.

Then it's yours: **tap** a channel to recall it (the QMX retunes instantly), **tap an empty slot** to store the current frequency/mode, **long-press** to edit, **drag** a channel to reorganise the grid, and **drag onto the wastebin** (bottom-right cell) to delete it.

Monitor from another room — or operate remotely. Once WiFi is configured, open `http://<tab5-ip>` in any browser for a full-featured live panadapter with mouse-driven tuning, band/mode controls, and ADIF log download. The IP is shown in the bottom status bar. See Web UI.

Step 6 — FT8 (if that's your thing)

Swipe in from the **left edge** to switch to the FT8 screen. The Tab5 starts decoding 15-second slots immediately — no PC, no WSJT-X required.

1. Tap the **Preset** button and pick your band's conventional FT8 dial frequency (e.g. 14.074 MHz for 20 m).
2. Watch the decode list fill. CQ stations appear at the top sorted by SNR; exchanges and replies below.
3. To reply to a station: **hold your finger on their row** for ~250 ms. A dim highlight appears after ~80 ms so you can see which row you're targeting before the gate fires. Lift — a confirmation modal shows the exact message before anything is armed.
4. Tap **Auto Pounce** to hand the full QSO to the auto-engine (works through report → RR73 → 73 with patient retry), or **Transmit** for a single manual message.
5. To call CQ: tap **Call CQ**. The engine picks a clear audio slot, fires CQ, automatically answers the first caller, runs the full exchange, logs the QSO, and resumes calling CQ.

Every completed QSO is written to an ADIF log downloadable from the web UI. See FT8 Transmit for the full picture, including ARRL Field Day exchange mode and a no-radio-keyed practice/simulation mode.

Step 7 — Something not working?

Spectrum flat, top bar shows `---`:

- Check the cable between Tab5 and QMX — almost always a charge-only cable.
- Make sure the QMX is powered on and not stuck in its own bootloader.
- Power cycle in order: Tab5 first, then QMX.

For anything else:

1. The diagnostic log is **always on** — nothing to enable.
2. Reproduce the problem (let it run a minute; power-cycle the QMX if the issue is about connection).

3. Grab the log:
    - **Over WiFi:** browse to `http://<tab5-ip>/api/log` or click **Diag** ↓ in the web UI bottom bar — downloads `qmx-log.txt`. After a reboot/power-loss, **Diag(saved)** ↓ (`/api/log/saved`) has the copy persisted to flash from before the reboot.
    - **microSD:** if a card is inserted, the log is mirrored continuously to `/qmx-panadapter/qmx-log.txt` on the card (alongside the always-on flash-persisted copy).
    - **Over USB (no WiFi needed):** capture the serial console with `tools/capture_serial_log.ps1`.
4. Open an issue and attach the log. It includes Tab5 and QMX firmware versions plus every CAT command exchanged — usually enough to pinpoint the problem immediately.

---

2. Panadapter

The panadapter is your primary view — a real-time spectrum analyser and waterfall for the QMX.

2.1. Layout

```
┌──────────────────────────────────────────────────────────────┐
│ 14.074 │ USB │ 2.5 kHz │ S7 │ 1.0x          [⚙️ settings] │  Top bar (60px)
├──────────────────────────────────────────────────────────────┤
│                     /\                                        │  Spectrum (200px)
│            ___/\_/  \___                                      │  Green trace,
│       /\/                 \                                    │   amber VFO,
│ ┼──────┼─────────────────────────────┼                         │   grey passband
├ 14.0  14.05  14.10  14.15  14.20 ┤  Frequency axis    │
│                                   │   (18px)           │
│ ░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░ │  Waterfall        │
│ ░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░ │  (412px)          │
│ ░░░░░███░░░░░░░░░░░░░░███░░░░░░░░░ │  Newest at top   │
│ ░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░ │  SDR gradient     │
├──────────────────────────────────────────────────────────────┤
│ Battery 95%  FPS: 30   SD  v1.1.0   Waterfall          │  Status bar (30px)
└──────────────────────────────────────────────────────────────┘
```

Status bar indicators:

Item	Meaning
Battery %	Current battery charge level
FPS	Render frame rate (target 30 fps)
SD (green dot)	A microSD card is mounted and being mirrored to `/qmx-panadapter/` on the card — a full grab-and-go station backup (see Settings)
`UTC(GPS)` / `UTC(NTP)` / `UTC(FT8)` / `UTC(FT4)`	The time source **currently** in charge — GPS-disciplined QMX, WiFi/SNTP, or FT8/FT4 offline sync (RTC / manual show as `UTC(RTC)` / `UTC(MAN)`)
Firmware version	Currently flashed firmware version
Waterfall label	Active colour map name

2.2. Touch Controls

Tap to Tune

Tap anywhere on the spectrum or waterfall to jump to that frequency. The panadapter snaps to a frequency grid (resolution depends on zoom level) for easy tuning.

- **Zoom 1.0x** — 10 kHz snap
- **Zoom 2.0x** — 5 kHz snap
- **Zoom 4.0x** — 2.5 kHz snap
- **Zoom 8.0x** — 1 kHz snap

Pinch to Zoom

Use two fingers to pinch in (zoom out) or pinch out (zoom in). The display centers on the passband width — in USB/LSB modes, the passband center stays on screen even when the VFO is off to the side.

Pan (One-Finger Drag)

Drag horizontally with one finger to scroll the spectrum left/right. Release to tune to the new center frequency.

Band-Plan Strip

Along the bottom of the screen — just above the status bar, below the waterfall — is a thin coloured strip showing where you are within the band at a glance. Each colour zone marks the conventional segment (CW, Digi, Phone) for your selected region (auto / ITU Region 1/2/3, set in settings).

The **visible-span block** inside the strip shows the exact portion of the band the spectrum and waterfall are currently displaying — it narrows as you zoom in and shifts as you pan.

A **passband sub-block** inside the visible-span block mirrors the current filter width at band scale (grey tint), so you can see how your passband sits within the CW/Digi/Phone zones.

Tune directly from the strip:

- **Tap** anywhere on the strip to jump to that frequency
- **Drag** to scrub along the band — the frequency label updates live and the QMX retunes on release

Drag from the bottom bar too. The visible-span block acts as a slider handle that reaches *below* the thin strip: touch on or just under it — anywhere along the bottom status bar — and **drag sideways** to scrub the band, exactly like dragging the strip itself. This gives you a much taller grab target. It coexists with the memory-picker gesture on the same row: a **sideways** drag retunes the band-plan, while a **vertical up-swipe** still opens the memory picker.

The strip updates live as you zoom, pan, or change bands.

Memory Channels

Swipe ↑ from the bottom edge to open the memory picker — a 4×8 grid of 32 channels, each free to hold any frequency/mode (not tied to a band). Each channel stores:

- Frequency
- Mode (USB, LSB, CW, DiGi)
- Label (your own name for the channel)

Each button shows the mode colour (CW = green, DiGi = teal, USB = brick red, LSB = purple) so you can tell mode at a glance without reading the label.

Recall: Tap a filled channel to immediately retune to it.

Tap an empty slot to create a new channel there directly — the frequency keypad opens straight away.

Long-press + drag a filled channel to move it to a different empty slot — no editing needed, the data follows your finger. Release on an empty slot to drop.

Long-press in place to open the editor (change name, frequency, or mode).

Entering a frequency outside the legal amateur band edges is rejected immediately with an error — the pad stays open so you can correct it without starting over.

2.3. Top Bar

Tap any item to open its selector:

Item	Purpose
14.074	Frequency. Tap to open the frequency keypad.
USB	Mode. Tap to cycle USB, LSB, CW, DiGi.
2.5 kHz	Bandwidth (SSB only). Tap to choose 2.5, 2.7, 2.9, or 3.2 kHz.
S7	S-meter. Shows received signal strength. Tap to reset peak hold.
1.0x	Zoom level. Tap to choose preset (1x, 2x, 4x, 8x) or custom.

2.4. Spectrum & Waterfall

The **spectrum** shows real-time signal strength (green curve) across the tuned band. The **waterfall** shows a rolling history of that spectrum, with colour indicating signal strength (SDR gradient: blue → green → yellow → red).

Flat Spectrum Mode (in settings) shows the dB scale relative to a per-bin noise floor, so even weak signals pop above the baseline. Normal mode shows absolute dBm (referenced to a -73 dBm S9 mark).

Waterfall Controls (in settings):

- **Black level** — how far above noise-floor to go black (default 9 dB)
- **Contrast** — dB span of the colour ramp (default 45 dB)
- **Adaptive floor** — blend between per-bin and global noise floor (default 100%)
- **FFT window** — Blackman-Harris, Hann, or Nuttall (default Blackman-Harris)

2.5. Display & Buffer Features

Spectrum Buffer Clear

When you **switch modes** (Panadapter ↔ FT8 ↔ FT4) or **change bands**, the panadapter automatically **clears the waterfall and resets the spectrum baseline**. This prevents stale signals from interfering with your new band or mode view.

- Waterfall clears completely (starts fresh)
- Noise floor recalibrates
- New baseline takes ~1 second to establish

This happens transparently — you'll notice the waterfall momentarily clear and re-initialize, then populate with real-time data from the new band/mode.

Flat Spectrum Mode

Toggle in settings. Shows dB scale relative to a **per-bin noise floor** (not absolute dBm), so weak signals stand out above the baseline even on a noisy band.

- **Normal mode**: absolute dBm referenced to S9 = -73 dBm
- **Flat mode**: relative dB above each bin's noise floor

2.6. Frequency Keypad

Tap the **frequency** on the top bar to open the keypad. Enter frequency in MHz format:

- **14.074** for 14.074 MHz
- **1.832** for 1.832 MHz
- **.100** for relative tune (±100 Hz from current)

Layout switches between **10-Key** (phone dial) and **Phone** (QWERTY) via a toggle. Choose whichever is faster for you — the preference persists.

Resize the keypad: Pinch it or swipe up/down on it to toggle between the normal and a compact layout. Your choice is remembered across reboots. The compact layout reveals more of the spectrum behind it.

Reposition the keypad: Drag it by the **"Enter freq"** title label (the only non-button area at the top of the panel) to move it anywhere on screen, clamped to stay fully visible. The position is remembered so it reopens exactly where you left it.

The background behind the keypad is intentionally semi-transparent (40% dim) so the spectrum and waterfall stay visible while you're entering a frequency.

Cancel and Enter are the only exits — tapping outside the keypad does nothing, so an accidental background touch cannot silently discard what you were typing.

2.7. Memory Channels

32 memory channels (4×8 grid), free to hold any frequency/mode — not tied to a band. Each stores frequency, mode, and name. Swipe up from the bottom edge to open the memory picker.

Mode is shown in colour on each button — CW (green), DiGi (teal), USB (brick red), LSB (purple) — so the grid is scannable at a glance without reading every label.

To recall a channel: Tap it. The QMX retunes immediately.

To create a new channel (empty slot): Tap the empty slot — the frequency keypad opens directly. No long-press needed.

To edit an existing channel: Long-press it in place. Change the name, frequency, or mode, then tap **Save**.

To move a channel to a different slot: Long-press and drag it to any empty slot. Release to drop. Only the data moves — the grid positions stay fixed.

To save the current QMX frequency/mode to a slot:

1. Swipe ↑ to open the memory picker
2. Long-press the slot you want to overwrite
3. The current frequency/mode pre-fills the editor — adjust the label if you like, then tap **Save**

To delete a channel — drag it onto the wastebin: the bottom-right cell (channel 32) is a **wastebin**. Long-press any filled channel and drag it onto the wastebin to delete it (it fades out). This is the way to clear a slot you no longer want.

Example channels on first use. A brand-new device ships with a few example channels already filled in (rather than 32 blank cells), so the grid is explorable straight away. These only seed empty slots — they never overwrite anything you've saved.

First-run tour. The very first time you open the memory picker, a brief (~10-second) one-time animation shows that channels can be **dragged** to move them and **dropped on the wastebin** to delete them. It plays once and never again.

Out-of-band frequencies are rejected immediately — if you enter a frequency outside a recognised amateur band, the keypad stays open so you can correct it rather than silently saving a wrong value.

2.8. Band Presets

Tap the **band name** on the top bar to switch between configured bands. The band selector shows:

- **160m, 80m, 60m, 40m, 30m, 20m, 17m, 15m, 12m, 10m, 6m** — standard bands (whichever your radio has configured)
- **11m** — the CB segment (26.9–27.5 MHz), shown on QMX+ radios that expose it
- **Custom** — user-added bands (via settings)

On radios with many configured bands (QMX+), the picker lays the bands out in **two side-by-side columns** so every band is visible without scrolling. Selecting a band jumps to the **nearest** band centre, so closely-spaced bands like 10 m and 11 m are no longer confused for one another.

Switching bands **remembers the last frequency you visited on each band**, so you can flip between 20m and 40m without losing your place.

2.9. Zoom & Pan

The zoom level is displayed on the top bar (e.g., **2.0x**). Tap it to choose a preset:

- **1.0x** — full 4 MHz span (standard panadapter view)
- **2.0x** — 2 MHz span (half-band view)
- **4.0x** — 1 MHz span (detailed search)
- **8.0x** — 500 kHz span (CW pile-up detail)
- **Custom** — (coming in a future release)

At zoom levels above 1.0x, the display **pans to keep the passband centered** when you change mode or bandwidth, so the active receive area stays on screen.

2.10. S-Meter

The S-meter (top-right of the top bar) shows received signal strength on a 0–68 scale:

- **S1 to S9** — standard S-units (-130 to -73 dBm)
- **+10 to +20** — above S9 in 10 dB steps

Tap the S-meter to toggle peak-hold mode (shows the strongest signal heard in the last ~5 seconds). Tap again to reset.

2.11. Settings Drawer

Swipe ← from the right edge to open the settings drawer. Common panadapter controls:

- **IQ Balance** — adaptive I/Q correction (usually on)
- **Flat Spectrum** — normalise to per-bin noise floor
- **Waterfall** — black level, contrast, adaptive floor, FFT window
- **WiFi** — on/off, SSID, password
- **Time Sync** — SNTP, manual time set, FT8-derived sync
- **Display** — 180° flip, brightness
- **About** — firmware version, reset to defaults

Next: Learn about FT8 Receive or Web UI.

Spectrum and waterfall

The display is divided into a **60 px top bar**, a **200 px spectrum** (green curve with dim fill), a **32 px frequency axis**, a **370 px waterfall**, a **22 px band-plan strip**, and a **36 px bottom bar**. The full visible span is 48 kHz centred on the QMX VFO.

The **spectrum** shows signal power in dBm (default range −130 to −30 dBm). Each frame is smoothed per-bin with an exponential moving average (EMA $\alpha = 0.4$ by default, adjustable in the drawer), balancing visual stability against snappy response to CW signals and SSB attack transients.

The **frequency axis** shows absolute MHz labels centred on the QMX VFO, refreshed on every CAT frequency update. At high zoom levels the labels resolve to kHz or Hz precision.

Band-plan strip. A thin coloured bar directly above the bottom status bar (below the waterfall) shows the coarse CW / Digi / Phone segments of the current band, with a marker for where the VFO sits within them. Pick which IARU region it reflects in the settings drawer → **Band-plan region** (Auto from your grid square, or a fixed Region 1 / 2 / 3).

The **waterfall** runs newest row at the top in a thermal SDR palette (black → dark blue → teal → green → yellow → red). Four colour maps are available in the settings drawer: Thermal, Viridis, Turbo, and Grayscale.

Waterfall floor tracking. The waterfall's black level tracks the band noise automatically — a running median sampled only from bins inside the passband, EMA-smoothed — so the background colour follows conditions rather than a fixed anchor. Bins outside the passband run darker and are excluded from the floor calculation so they don't wash out dim in-band signals.

Flat-spectrum mode (toggle in the settings drawer, persisted to NVS) switches both spectrum and waterfall to a per-bin adaptive display: each bin renders as dB above its own running noise floor. Real signals — including weak CW tones — stand out sharply against a calm baseline. This is the recommended mode on noisy bands. The dB range sliders have no effect in flat mode; the axis shows relative dB above floor.

Touch-to-tune

Tap anywhere on the spectrum or waterfall to place the cyan tune cursor. Drag and the cursor snaps grid-point to grid-point — you can see exactly which frequency will be tuned before you lift. The snap grid is **mode-aware**:

Tune vs. pan — how your finger is read. A quick horizontal swipe (more than ~70 px within 250 ms) pans the view instead of tuning — see Zoom and pan. A slower touch-and-hold (250 ms without that much movement) locks into tune mode, after which dragging moves the snap cursor as described below. In short: swipe fast to pan, hold-then-drag to tune.

Mode	Snap step
CW / CW-R	10 Hz
USB / LSB	250 Hz
FT8 / DiGi / RTTY	500 Hz
AM	1 kHz

The grid is anchored to absolute frequency (e.g. …200 / 300 / 400 Hz), not to your touch start point, so the cursor always lands on the same set of grid points regardless of where your finger first touches.

A floating frequency tooltip above the cursor shows the target frequency in real time while dragging. Lift → CAT `FA` command is sent; QMX retunes; spectrum re-centres.

Taps always tune to exactly where you touched (snapped to the mode-aware grid above). The old **Snap to signal** option — which hunted for the strongest bin near your tap — was removed in v0.19.4; predictable tuning won.

Passband indicator. Two grey vertical lines mark your current filter edges. A faint coloured tint fills the passband. The amber VFO marker shows where the QMX is tuned; in CW mode it sits at dial + CW pitch offset so it marks the actual received tone frequency, not the suppressed carrier.

Zoom and pan

Gesture	Effect
One-finger fast horizontal swipe	Pan/"stroll" the view — retunes to the new centre frequency on release
Pinch (two fingers)	Zoom ×1.0 – ×24.0
Two-finger drag	Pan the zoomed window
Double-tap	Reset zoom and pan to ×1.0 / centred
Top-bar Zoom → tap	Zoom preset: ×1 / ×2 / ×4 / ×8 / ×16 / ×24

One-finger pan (stroll). A fast horizontal swipe — more than ~70 px of movement within the first 250 ms of touching down — slides the spectrum and waterfall under your finger in real time, with a live frequency tooltip, and retunes to wherever you release. This works at any zoom level alongside the two-finger pinch/pan above; it's the quickest way to slide along a band without lifting into a deliberate tune-drag.

At zoom > ×1 the view **centres on the passband** (not the VFO dial), which matters for USB/LSB where the passband sits offset from the carrier. The passband lines and frequency axis track correctly at all zoom levels. Zoom level is persisted to NVS and restores with full zoom-FFT resolution on the next boot.

S-meter

The Signal field in the top bar is a visual tick-scale bar labelled S1, S3, S5, S7, S9, +10, +20 with a moving green bar beneath. It is driven by the peak dBm in a ±64-bin window centred on the IF-shifted VFO bin — the actual signal under your cursor, not the DC/LO artefact at bin 0.

S-unit mapping: S9 = −73 dBm; 6 dB per S-unit below S9; 1 dB per unit above S9. The meter stays live during FT8 capture.

Memory channels

Swipe up from the bottom edge (or tap the bottom grip handle) to open the 4×8 memory channel grid (32 slots, NVS-persisted).

Action	How
Recall	Tap a slot — QMX retunes to stored frequency and mode
Create	Tap an empty slot — the frequency/mode picker opens directly
Edit	Long-press an occupied slot
Move	Long-press and drag a filled slot onto an empty one — the data follows your finger
Delete	Drag a filled slot onto the **wastebin** (bottom-right cell, channel 32) — it fades out

Memory slots show the label in large text and mode + frequency (dimmed) below. The frequency/mode picker pre-fills the current VFO and lets you edit both before naming.

A new device ships with a few **example channels** already filled (rather than 32 blank cells), seeded only into empty slots so they never overwrite anything you've saved. The **first time** you open the picker, a brief one-time (~10 s) tour demonstrates the drag-to-move and drag-to-wastebin gestures, then never plays again.

Settings drawer

Open by swiping in from the right edge, or tapping the right grip handle. The drawer is scrollable.

Controls appear top to bottom in this order:

Control	What it does
Flip 180°	Inverts the whole display and touch axes for upside-down mounting; centred checkbox so it isn't hit by accident, persisted
IQ Balance	Toggle adaptive I/Q image correction; re-enabling resets the estimator
Flat spectrum	Toggle flat/absolute display mode, persisted
Presets	HF Normal / HF DX / Strong Sig — sets dB range and smoothing in one tap
WiFi	Opens credential modal; **WiFi initiated** checkbox enables/disables WiFi entirely
Identity	Callsign + Maidenhead grid (required for FT8/FT4 TX; also drives KM/BRG columns)
Band-plan region	Auto (from your grid square) / Region 1 (EU/AF) / Region 2 (Americas) / Region 3 (Asia/Pac) — drives the band-plan strip; sits right under Identity since Auto derives from your grid
dB Range	Min and Max sliders (dBm)
Smoothing	EMA alpha 0.05–1.00
CW	CW sidetone centre, 600–800 Hz; touch-to-tune in CW mode snaps to this offset, persisted
IF calibration	±100 Hz trim for per-unit LO variance (see Per-unit IF calibration)
Display	Brightness, 10–100%, persisted

Control	What it does
Battery charge limit	Optionally stop charging at a set percentage (default 80%) to reduce long-term pack wear; charging resumes automatically if the level later drops well below it (5% hysteresis). The displayed charge % now also compensates for the voltage rise while charging, so it no longer jumps around when plugged in
Waterfall colour map	Thermal / Viridis / Turbo / Grayscale, persisted
Waterfall	Black level, Contrast, Adaptive floor blend, and FFT window — see Waterfall colourisation below

The drawer was decluttered in v0.19.4: the **Snap to signal** and **FT8 sync lines** toggles were removed (taps now always tune where you touch), and **Band-plan region** moved up next to Identity.

The drawer shows a different subset while on the FT8 screen (Flip 180°, WiFi, Identity, Display — plus three FT8-only controls not shown above: **Distance in miles** for the decode list's KM/MI column, **Fast pounce (early decode)** — see below — and **FT8 Simulation Mode**, see FT8 Simulation mode).

Fast pounce (early decode) — on by default. Decodes surface ~1.8 s *before* the slot boundary (the way WSJT-X decodes in the dead-air gap), so replying to a fresh CQ can transmit in the very next slot instead of waiting a full 30 s cycle, and mid-QSO replies land on the beat. The trade-off: the capture window closes ~1.8 s early, so a station transmitting *late* in the slot can occasionally be clipped and missed. ⚠️ This feature has not yet been A/B-verified on a live band — if your decodes-per-slot drop noticeably with it on, turn it off here and please report your before/after numbers on the groups.io thread.

Waterfall colourisation

Four live, NVS-persisted sliders/dropdown at the bottom of the settings drawer fine-tune how the waterfall maps signal to colour — changes scroll in from the top as you drag:

Control	Range	Default	Effect
Black level	0–30 dB	9 dB	How far above each bin's own noise floor a signal needs to be before it lifts off black
Contrast	10–80 dB	45 dB	The dB span that fills the rest of the colour ramp above the black level
Adaptive floor	0–100%	100%	Blend between a per-bin noise floor (100%) and one global mean floor across the band (0%)
FFT window	Blackman-Harris / Hann (sharp) / Nuttall	Blackman-Harris	The FFT window function; Hann trades some sidelobe suppression for sharper peaks

---

3. Web UI

With the Tab5 on WiFi, open `http://<tab5-ip>` in any modern browser. The IP is shown in the bottom status bar on the Tab5.

The browser panadapter is a full-featured view in its own right — not just a window onto the Tab5. On a larger monitor you get more spectrum history, a bigger waterfall canvas, and mouse controls that are faster than touch for precise tuning. It shows live spectrum at ≈10 fps via WebSocket, full waterfall history (~50 s), the same thermal palette and floor maths, a graphical S-meter, and a top bar with Band / Mode / BW / Zoom controls. The bottom bar shows battery percentage + voltage, firmware version, a live UTC clock, and WiFi SSID + RSSI. To its right: download/upload buttons (ADIF, QRZ, eQSL — see QSO logging), **Diag ↓** for the diagnostic log, **Config ↓ / Config ↑** to back up / restore / edit all settings (see Config backup, restore & edit), and **Tab5Shot** which opens a live `/ss.bmp` screenshot in a new tab.

microSD file browser (new in v1.3.0). Files → SD Files in the bottom bar opens `http://<tab5-ip>/files` — browse the microSD card from any computer without pulling it: download your logs, config backups, and offline manual; upload files; delete. Card access is coordinated with the WiFi link the same way the automatic backup is, so it's safe to use mid-session.

Whole-band plan strip & adjustable split (new in v0.20.0). Along the bottom of the browser view (above the status bar), a colour-coded CW/Digi/Phone strip spans the entire band with a draggable "visible window" (drag or tap to retune) and a VFO marker, mirroring the Tab5's own strip. Drag the divider between the spectrum and the waterfall to give either more room — the split is remembered in the browser. **Tab5Shot** now captures any open pop-up (band/mode dropdown) too, and the frequency keypad is draggable with a standard 10-key layout.

In FT8/FT4 mode the live stream pauses (new in v0.20.0). The browser stops streaming the spectrum/waterfall and shows a notice plus the log and upload controls instead. While you're operating digital modes the stream would compete with the on-device decoder and the WiFi link, so pausing it keeps FT8 decoding and WiFi noticeably steadier. Switch the Tab5 back to Panadapter mode and the stream resumes automatically.

Click or drag to tune. Click or drag on the spectrum or waterfall — a cyan cursor appears with a live frequency readout and commits on release.

Mouse-wheel to pan or tune. Rolling the mouse wheel over the spectrum or waterfall **pans** the view when zoomed in, letting you survey the band without touching the Tab5. At zoom ×1 the wheel **tunes** with mode-aware snap (CW 10 Hz, SSB 500 Hz, DiGi 100 Hz, AM 1 kHz).

Band / Mode / BW / Zoom dropdown pills in the top bar send commands to the QMX via `/api/cmd` — the same effect as using the Tab5 top bar.

Zoom sync. The browser renders the same zoomed window as the Tab5.

ADIF log download. Once you have logged at least one completed FT8 QSO, a **"N QSOs ↓"** link appears in the bottom bar of the web UI. Clicking it downloads your `qso.adi` file directly. The link is only shown when the log contains data — it disappears after clearing the log with `/api/adif/clear`.

QRZ / eQSL upload. Two more buttons appear alongside the ADIF link once you have logged QSOs — see QSO logging for the full picture.

Config backup, restore & edit

The **Config ↓ / Config ↑** buttons in the bottom bar let you save every setting to a plain text file, edit it on your PC, restore it, or share parts of it.

- **Config ↓** downloads `qmx-config.txt` — a human-readable INI-style file with everything the Tab5 remembers, grouped into sections:
  - `[settings]` — callsign, grid, WiFi SSID/password, CW pitch, IF trim, IQ balance, flat-spectrum, zoom, colour map, brightness, dB range, EMA, GPS toggle, keypad layout, QRZ key, eQSL user/pass.
  - `[cq]` — the three FT8 CQ-message presets and which is active.
  - `[ft8_filters]` — the CQ-run include/exclude filter terms and toggles.
  - `[memories]` — the 32 memory channels, one per line: `slot = freq_hz, mode, label`.
- **Config ↑** picks an edited file and **merges** it back: only the keys and sections present in the file are changed — everything else is left as-is. Unknown keys are ignored (so a file from a newer firmware still loads). Memory channels apply immediately; other settings take effect after a restart.

Three things you can do with it:

1. **Back up before a clean flash.** Hit **Config ↓** first, do the clean/erase flash, then **Config ↑** to restore everything in seconds — instead of re-typing it all on glass.
2. **Edit in a text editor instead of on the touchscreen.** Faster for callsign, grid, CQ messages, or punching in a batch of memory channels.
3. **Share a band plan.** Copy just the `[memories]` section into a message — another operator pastes it into their file and uploads it to get the same channels (their other settings untouched).

> The downloaded file contains your **WiFi password** and **QRZ/eQSL credentials in clear text** (so it works as a full backup). Keep it private; strip those lines before sharing a file with anyone.

Endpoints

Endpoint	Method	Returns
`/`	GET	Browser panadapter (HTML)
`/api/status`	GET	JSON status (see below)
`/api/cmd`	POST	Send Band/Mode/BW/Zoom commands
`/api/log`	GET	Diagnostic log download (`qmx-log.txt`)
`/api/log/saved`	GET	Flash-persisted diagnostic log from before the last reboot/power-off
`/api/adif`	GET	ADIF QSO log download (`qso.adi`)
`/api/adif/clear`	GET	Wipe ADIF log and worked-call cache
`/api/qrz_key`	POST	Save QRZ Logbook API key (body = raw key text)
`/api/qrz_upload`	POST	Upload pending QSOs to QRZ Logbook; returns `{uploaded, failed, error}`
`/api/eqsl_creds`	POST	Save eQSL username/password (JSON body `{"user","pswd"}`)
`/api/eqsl_upload`	POST	Upload pending QSOs to eQSL (batched); returns `{uploaded, failed, error}`
`/api/config`	GET	Download all settings + memory channels as editable INI (`qmx-config.txt`)
`/api/config`	POST	Upload an INI config; merges into NVS; returns `{applied}`
`/ss.bmp`	GET	1280×720 RGB565 BMP screenshot
`/ws`	WS	Binary spectrum stream (~10 fps)

`/api/status`

```
{
  "battery":     { "level": 100, "mv": 8320, "charging": true },
  "wifi":        { "ssid": "MyNet", "rssi": -44, "ip": "192.168.1.213" },
  "freq_hz":     14074000,
  "mode":        "USB",
  "band":        "20m",
  "passband_hz": 2700,
  "signal_dbm":  -87.4,
  "zoom":        1.0,
  "pan_bins":    0,
  "flat_mode":   false,
  "utc_epoch":   1750000000,
  "qso_count":   12,
  "qrz_key_set": false,
  "eqsl_creds_set": false,
  "tab5_fw":     "v0.18.8",
  "qmx_fw":      "1_03_002QMX"
}
```

Polled at 1 Hz by the landing page; safe to consume from monitoring scripts, home automation, etc. `qmx_fw` is read from the QMX via the `VN;` CAT command at link-up (empty until the radio responds). `signal_dbm` is the peak dBm around the IF-shifted VFO bin (null if DSP has no data yet). `qrz_key_set` / `eqsl_creds_set` tell the web UI whether to prompt for credentials before the next upload.

`/ws` — binary spectrum WebSocket

Single-client endpoint. Each binary frame is 1026 bytes:

Bytes	Meaning
`0`	Frame type: `0x01` = spectrum
`1`	Zoom decimation factor (1 = base 1024-bin spectrum, >1 = zoom-FFT with that decimation)
`2–1025`	1024 unsigned bytes, one per bin, quantised −130 dBm (0) to −30 dBm (255)

Bins are pre-shifted server-side so byte 2 is the leftmost screen pixel. Push rate ≈10 fps.

Screenshots

```
Invoke-WebRequest -Uri "http://192.168.1.213/ss.bmp" -OutFile screenshot.bmp
```

---

4. FT8 Receive

The panadapter includes **on-device FT8 and FT4 decoders** with real-time spectrum waterfall and a live list of heard stations.

4.1. FT8 & FT4 View

Swipe → from the left edge to toggle to FT8/FT4 view. The same decode list and waterfall work for both modes — switch modes via the **Preset** dropdown in the left pane (top).

FT4 notes:

- Decodes refresh roughly twice as fast as FT8 because slots are 7.5 seconds
- Slot countdown shows 7.5 s instead of 15 s
- All filtering, priority, and auto-reply features work identically in FT4
- Time-sync from decoded signal timing works in both FT4 and FT8 (the bottom-bar clock shows `UTC(FT4)` when synced from an FT4 decode)
- **v0.19.4 made FT4 reliable**: a memory-placement fault previously killed decoding on every other slot (FT8's longer slots mostly hid it), the EVEN/ODD slot markers were on the wrong 15-second grid, and the slot clock could visibly jump between slots — all fixed; if FT4 seemed deaf or erratic on an earlier version, update the firmware

You'll see:

- **Decode list** (left pane) — all heard stations, sortable by signal strength, distance, or newest
- **Waterfall** (right pane) — same real-time spectrum as panadapter mode
- **Call CQ button** — start a CQ (requires QMX on the air)
- **Filter button** — include/exclude stations, prioritise by SNR/distance, show only CQ callers

4.2. Decode List

The list shows every decoded FT8 message:

Column	Meaning
Callsign	Their call, or `CQ` if they're calling CQ
Report	SNR (signal-to-noise) or grid square they're sending you
Grid	Their grid square (estimated from signal geometry)
Time	When the message was decoded (slot boundary)

Own call highlight — your callsign is shown in **inverted colours** (red fill, white text) so you spot replies to you instantly.

Stale entries — the list is a live picture of who's on frequency now; stations not heard again within **2 minutes** drop off automatically.

During your own CQ run — stations transmitting on *your* transmit slot (which you can't hear while transmitting) are hidden from the list and return when the run ends; other stations' CQ rows are hidden during a run as before.

Nonstandard callsigns — special-event and compound calls now resolve to the full callsign instead of showing `<...>`, once the station has been heard in full.

4.3. Filtering & Priority

Tap the **Filter** button to open the filter modal:

- **Include callsigns** — only show these calls (space or comma-separated)
- **Include callsigns containing** — show any call with these substrings (e.g., `/P` for portable)
- **Exclude callsigns** — hide these calls
- **Exclude if contains** — hide any call matching these substrings
- **Exclude worked before** — skip calls you've already logged QSOs with
- **Show only CQ callers** — hide replies, show only CQ messages
- **Skip TX1** — when you pounce a station, open with your signal report straight away instead of the grid exchange, for a quicker QSO (falls back to the normal grid exchange if the station has dropped out of the decode list)
- **Auto-reply priority** — Strongest SNR, Weakest SNR, Most distant grid
- **Robot mode** — auto-answer CQ (disabled by default; see FT8 Transmit)

All filters persist across sessions. Toggle any filter on/off to enable or disable it without erasing the criteria.

4.4. Decoding Performance

On-device decoding runs at:

- **FT8:** ~15–20 QSOs per 15-second slot
- **FT4:** ~15–20 QSOs per 7.5-second slot (faster decode cadence, same per-slot throughput)

This is sufficient for casual operation but not a full WSJT-X-equivalent decoder.

Why not faster?

- **Single CPU (core 0)** decodes ~1–2 QSOs/s (LDPC is computationally heavy)
- **Real-time FFT** (waterfall) competes for CPU cycles
- **USB audio streaming** needs constant servicing

FT4 advantage: because FT4 slots are half as long (7.5 s vs 15 s), you get refresh feedback roughly twice as fast, which feels snappier on a crowded band even though the per-slot decode count is the same.

The panadapter prioritises **receive latency** (decode every 15 s, not accumulate) over volume.

4.5. Real-Time Waterfall

Unlike WSJT-X (which buffers a whole 15-second capture), the Tab5 **builds the waterfall on-the-fly** as audio arrives. You see CW and other traffic in real time, and FT8 stations appear instantly as their symbol blocks are decoded.

The waterfall also shows **non-FT8 activity**:

- CW (carrier lines)
- SSB (broader vertical smears)
- RTTY (FSK sidebands)
- Interference (wideband noise)

4.6. Tap to Reply

Tap any FT8 row in the decode list to **prepare a reply**:

1. A confirmation modal pops up showing your message

2. You can review the target callsign, grid, and SNR
3. Tap **Transmit** to send, or **Cancel** to abort

The reply always follows FT8 protocol (correct parity, proper message sequence) — you can't accidentally send a garbled or out-of-order message.

Pile-up list. If a station answers you while you're busy with another contact, they used to disappear from the list once they stopped transmitting. They're now remembered in a **Pileup** list — the **ADIF-log** button on the FT8 screen turns into a **Pileup** button (a different colour) whenever callers are waiting. Tap it to see who's called you, tap a station to work them, or tap the ✕ to dismiss one. See FT8 Transmit for the full workflow.

4.7. Auto-Reply (Robot Mode)

⚠️ **Experimental** — enabled via a checkbox in the Filter modal.

When robot mode is on, the Tab5 **automatically replies to CQ** without waiting for you to tap. It scans each FT8 slot for CQ callers matching your filters, picks the highest-priority station, and sends a full QSO exchange (TX1 → wait for report → TX2 → wait for RR73 → TX3). Everything is logged to ADIF.

Important: Robot mode **keys the QMX for real** — your signal goes on the air. Never leave it unattended unless you're confident in your filters and your station is in a safe state.

4.8. Search & Window

The decoder uses an **FFT-based search window** to find candidate tone blocks. The search is tuned for standard FT8 tone spacing (6.25 Hz per tone).

Frequency stability: The QMX's USB audio clock isn't bit-exact 48 kHz, so decodes slide slowly over time. The panadapter **re-locks to UTC boundaries every 15 seconds** (the slot boundary), preventing long-term drift.

4.9. Time Sync for FT8

FT8 requires **UTC time accurate to ±1 second** for slot synchronisation. The panadapter gets time from:

1. **WiFi + SNTP** (if available) — most accurate, syncs every ~hour
2. **Tab5 RTC** (if set) — accurate to ±1 min, no internet needed (POTA)
3. **QMX GPS** (if QMX has GPS) — re-syncs the RTC every 5 minutes
4. **Manual** — you can set time via the settings drawer

See Time Sync for details.

Next: Learn about FT8 Transmit or go back to Panadapter.

Live FT8 reception on 20 m (v0.15.7). Left pane: MODE / Preset freq / UTC / slot countdown with parity and progress bar / TX parity preference / operator identity / Call CQ / decode summary. Right pane: scrollable decode list.

Swipe in from the **left edge** to switch to the FT8 screen. The Tab5 decodes 15-second FT8 slots directly on the ESP32-P4 — no PC, no WSJT-X.

Prerequisites

- **WiFi configured** — recommended for accurate UTC time. Without it the Tab5 falls back to its supercap RTC or the QMX clock. See Time sync.
- **Callsign and grid** set in the drawer → **Identity** if you want distance/bearing columns or plan to transmit.

The FT8 screen

Left pane: MODE label · **Preset** frequency button (tap to pick a band's conventional FT8 dial frequency) · UTC clock · 15-second slot countdown with current parity (EVEN blue / ODD amber) and a gliding progress bar · TX: EVEN / TX: ODD parity preference buttons · Active station count · **Call CQ** button · last-slot decode summary.

Right pane — decode list:

Column	Content
SL	Slot parity: **E** (blue, :00/:30) or **O** (amber, :15/:45)
CALL	Extracted callsign
MESSAGE	Full decoded FT8 message text
COUNTRY	DXCC entity from prefix lookup (~190 entities)
SNR	FFT-based estimate, colour-banded: green ≥0 / white −5..−1 / orange −15..−6 / grey <−15
KM	Great-circle distance from your grid
BRG	Bearing from your grid
HRD	Times decoded since last appearance

CQ calls always appear at the top sorted strongest-SNR first; all other rows follow by SNR descending.

Row colour scheme (CALL + MESSAGE columns): **white** for ordinary traffic, **green** for plain `CQ` calls, **dim grey** for callsigns already worked **on the current band** (a station you worked on 20 m still shows normally on 40 m — see Reply filter for hiding them entirely), and **inverted red fill + white text** for any message containing your own callsign — your highest-priority rows literally pop off the screen instead of relying on red-on-black text, which field testing found hard to read at a glance.

Live view. Stations not re-decoded within 60 seconds drop off the list automatically, even while the band is quiet. The count reads "Active: N" — who's on frequency *now*, not a cumulative history.

Performance

On a busy 20 m FT8 slot the decoder regularly yields 25–50 callsigns per slot. Both EVEN and ODD slots decode every cycle via a ping-pong dual-buffer architecture — a TX slot never causes the opposite parity to be skipped. Heap is stable across long sessions (≈39 KB internal RAM free, 25 MB PSRAM free during decoding).

---

5. FT8 Transmit

The panadapter **keys the QMX and transmits full FT8/FT4 QSOs** — reply to CQ, run your own CQ, auto-answer (robot mode), or conduct a full exchange.

5.1. FT8 vs FT4 — Which to Use?

Both modes transmit via the same CAT interface; the difference is **slot length and symbol rate**:

Feature	FT8	FT4
Slot length	15 seconds	7.5 seconds
Symbols transmitted	79	105
Symbol duration	160 ms	48 ms
QSO time (full exchange)	~90 sec (6 slots)	~45 sec (6 slots)
Use case	Weak signal, SOTA/POTA	Busy bands, contests
Preset picker	Panadapter mode → Freq dropdown	FT8 mode → Preset picker

Choose FT4 when:

- The band is crowded (faster exchanges, less QRM)
- You're in a contest (time is critical)
- You want quicker pileup resolution

Choose FT8 when:

- Signal is marginal (longer decode window tolerates more QRN)

- You're portable/POTA (less pressure to transmit fast)
- You prefer the relaxed pace

FT4 has full feature parity with FT8, including time-sync from decoded signals (fixed in v0.19.3 — the time-sync modal shows "FT4" when appropriate and the bottom-bar clock correctly reads `UTC(FT4)` while in FT4 mode).

5.2. Switching Between FT8 and FT4

In FT8 screen (left pane):

- Tap the **Preset** dropdown (shows current frequency/mode)
- Select an FT4 frequency to switch to FT4
- Select an FT8 frequency to switch back

In Panadapter screen:

- Tap the **Band preset dropdown** (top left)
- Some bands have both FT8 and FT4 frequencies listed
- Tap the FT4 variant to switch

Your **settings and decode list are sticky** — when you switch back to FT8 or FT4, the panadapter remembers your last frequency, bandwidth, and filter settings for each mode.

5.3. Modes of Transmission

5.0.1. Reply to a CQ

In FT8 view, tap a CQ row in the decode list. A confirmation modal appears:

```
┌──────────────────────────────┐
│ Confirm FT8 Transmission     │
│ ▲  K9ZZ EN52  -07  2138 Hz  │ ← nudge up
│ ►  W1AW FN31  -12  1406 Hz  │ ← selected row
│ ▼  SV1ENG JN37 -09 1869 Hz  │ ← nudge down
│                              │
│  TX1: W1AW OZ1LAV JO45      │
│                              │
│  [▲ Nudge up]  [▼ Nudge down]│
│  [Auto Pounce]    [Cancel]   │
│  [Transmit ●]                │
└──────────────────────────────┘
```

- **Transmit** (green) — send the reply on the next correct slot
- **Auto Pounce** (blue) — handle the entire exchange automatically (TX1 → wait for report → TX2 → wait for RR73 → TX3 73)
- **▲ / ▼ Nudge** — move the target to the row above or below, without closing the modal and redoing the selection gesture
- **Cancel** — abort

A quick tap is enough — you do not need to hold a row before releasing. Hold-and-drag still works for selecting across rows in a busy list.

The reply follows **correct FT8/FT4 parity** — if you're replying to an even slot, you transmit on the odd slot, and vice versa. For FT4 the countdown timer correctly counts down in 7.5-second slot increments, not 15-second FT8 ones.

Skip TX1 (faster pounce). With **Skip TX1** enabled in the Filter modal, pouncing a station opens with your signal report straight away instead of the grid exchange — saving one round trip. If the station has already dropped out of the decode list, it falls back to the normal grid exchange automatically.

Working a pile-up

When you call CQ (or work a busy run), more than one station may answer at once — and a caller who replies while you're mid-QSO with someone else used to vanish from the live decode list as soon as they stopped transmitting. They are now kept in a **Pileup** list so you don't lose them:

- Whenever one or more stations are waiting, the **ADIF-log** button on the FT8 screen becomes a **Pileup** button in a distinct colour, and reverts once the list empties.
- Tap **Pileup** to see everyone who has called you and isn't worked yet.
- Tap a station in the list to work them (the same confirmation modal as a decode-list tap), or tap the small ✕ to dismiss them without working them. Replies from the pile-up list open with a **signal report** directly — the correct opening for a station that has already called you.
- A station is removed from the list automatically once you start a QSO with them.

The pile-up tracker never transmits on its own — it only remembers callers; you choose who to work.

5.2. Call CQ

Tap the **Call CQ** button. A modal opens:

```
┌─────────────────────────────┐
│ CQ Preset:                  │
│ [○] 1. CQ OZ1LAV JO45       │
│ [○] 2. DX OZ1LAV JO45       │
│ [○] 3. POTA OZ1LAV JO45     │
│                             │
│ [Save Preset] [Transmit]    │
└─────────────────────────────┘
```

Choose a preset (or edit/save a new one), then tap **Transmit**. The QMX starts calling CQ and listening for replies.

CQ tone selection — the panadapter automatically picks a quiet frequency (6.25 Hz tone spacing, 200–2800 Hz audio range, no other stations nearby). If your chosen tone gets busy during a QSO, the panadapter **auto-relocates to the nearest clear frequency** to avoid stepping on other stations.

5.3. Auto-Reply (Robot Mode)

⚠️ **Requires explicit checkbox in the Filter modal** to enable.

When robot mode is on and you're idle, the panadapter scans every FT8 decode for CQ callers matching your filters, picks the highest-priority match (Strongest SNR, Weakest SNR, or Most distant grid — your choice), and starts a full QSO:

1. **TX1** — send your call, grid
2. **RX** — wait for their report
3. **TX2** — send their report + your call back
4. **RX** — wait for RR73 or 73
5. **TX3** — send 73 or RR73 (exchange complete)
6. **Log** — ADIF entry created, move to next CQ

Each QSO takes **~90 seconds** (6 FT8 slots × 15 s/slot). If a station doesn't respond in 4 slots, the panadapter times out and resumes scanning CQ.

Important: Robot mode **transmits on the air unattended**. Never enable it unless:

- Your station is physically attended
- Your filters are tested and correct
- Your antenna/SWR is known good

- Your QMX power is set appropriately

A permanent on-screen warning appears whenever robot mode is available.

5.4. FT8 Simulation Mode (Practice QSOs)

⚠️ **For practice only** — no real stations involved, radio never keys up.

Enable **FT8 Simulation Mode** in the settings drawer to practice full QSO exchanges with phantom stations (W1AW and K9ZZ) without transmitting on the air. Useful for:

- Learning the full FT8 exchange sequence
- Testing your setup without risking interference
- Verifying ADIF logging works correctly
- Practicing in a realistic environment

How it works:

- Two phantom stations periodically call CQ on their own schedule
- You can reply, they respond, and complete a full QSO
- Messages and timing are identical to real FT8 — the decoder runs actual GFSK synthesis
- Every simulated QSO logs to ADIF with the phantom callsign

Visual indicator: When simulation mode is active, a **10 px red border pulses around the entire screen** (breathing red frame). This is your visual reminder that you're in practice mode — if the red frame is gone, simulation is off and real stations are in play.

Important:

- The QMX is **never keyed** in simulation mode, no matter what
- QSOs are logged as real ADIF entries (you may want to delete practice contacts from your log afterward)
- All other features (panadapter, web UI, settings) work normally

5.5. ARRL Field Day Mode

During ARRL Field Day, enable **Field Day mode** in the settings:

- **Mode** — on/off
- **Class** — your transmitter class (1–2 kW)
- **Section** — your ARRL section

When FD mode is active:

- **CQ message changes** to `CQ FD` (replaces any other modifier)
- **Exchange message** includes your class and section (e.g., `5B WCF` = class 5, section WCF)
- **ADIF record** logs `CONTEST_ID=ARRL-FD`, your section, and their section

The full FD exchange is automatic — no manual mode switching needed.

5.6. Message Status

The FT8 left pane shows a **status label** indicating what's happening:

Status	Meaning
ACTIVE (red)	Transmitting right now
ARMED (amber)	Ready to transmit next slot
QSO Complete (green)	Exchange finished, logged to ADIF
QSO Timeout (orange)	No response after 4 slots, exchange aborted
(dim white)	Idle, no active QSO

Tap the status label to **abort** the current QSO (only works if ARMED or ACTIVE; once COMPLETE, it's logged).

Resume after timeout — if a QSO times out because the partner faded, and they come back within ~5 minutes calling you, the exchange **resumes where it left off** automatically (or tap their row to resume manually) instead of restarting from scratch.

5.7. Power & SWR Readout

After each transmit burst, the status bar briefly shows:

```
Last TX: 5.2W SWRx1.25 [N=79]
```

- **5.2W** — average output power measured via QMX `PC;` CAT command
- **SWRx1.25** — SWR measured via QMX `SW;` CAT command (only valid during TX)
- **[N=79]** — number of symbols transmitted
  - FT8: always 79 symbols (~12.7 s at 160 ms/symbol)
  - FT4: always 105 symbols (~5.0 s at 48 ms/symbol)

These readings are **informational only** — the panadapter does not enforce limits. Monitor your antenna system independently.

5.8. CQ Presets

You can save up to 3 custom CQ messages:

1. Tap **Call CQ**
2. Tap **Edit Presets** (or long-press the active preset)
3. Type your message: callsign, modifier (optional), grid
4. Tap **Save**

Presets persist across power cycles. Common modifiers:

- **DX** — calling DX only
- **POTA** — Parks on the Air activation
- **SOTA** — Summits on the Air activation
- *(blank)* — standard CQ

5.9. Frequency & Tone Control

When a CQ is active, the panadapter holds your **CQ tone** (6.25 Hz audio frequency) fixed across slots. If another station lands on your tone during an exchange, you'll see a warning ⚠ **FREQ BUSY** on the status label — but the QSO continues on the same tone (doesn't auto-relocate mid-exchange).

To tune to a different frequency while CQ is running:

1. Tap the **Freq** label on the top bar
2. Enter a new frequency
3. The CQ continues on the new frequency next slot

5.10. ADIF Logging

Every completed QSO is **automatically logged to ADIF** on the Tab5's internal storage:

- **Callsign**, grid, report (SNR), time (UTC)
- **Frequency**, mode (`FT8`)
- **Duration** (QSO start to finish)
- **Operator** (your callsign)

Download the log via the web UI (**QSO Logs** menu → **ADIF download** ↓) or Settings → ADIF Log.

5.11. Upload to QRZ, eQSL & LoTW

Via the web UI:

1. Open the **QSO Logs** menu in the bottom bar and click **QRZ upload** ↑, **eQSL upload** ↑, or **LoTW setup** (reads **LoTW** ↑ once configured)
2. Enter your API key (QRZ), username/password (eQSL), or run the guided certificate setup (LoTW) when first prompted — credentials are saved for future sessions
3. Click again to upload

See Web UI → LoTW Upload for the LoTW certificate setup details.

Logs are batched — each upload session records which QSOs have been sent, so re-running skips already-submitted entries.

Uploads work **while FT8 or FT4 is actively running** — the panadapter briefly steps the FFT and SD-archive activity aside for the duration of the HTTPS transfer, then resumes automatically. You typically miss one FT8 slot and won't notice. A progress result is shown once the upload completes.

5.12. Troubleshooting

"QMX not responding to TX command"

- Check USB cable (must be data cable, not charge-only)
- Verify QMX is in the correct mode (USB/LSB/CW/DiGi, not AM)
- Try a manual `TX;` command via the web UI's CAT tab

"⚠ FREQ BUSY warning, but no other station is there"

- The panadapter's tone-clash detector can be overly sensitive on a busy band
- Tap the status label to clear the warning and continue
- See Settings → FT8 Filters for tone-clash sensitivity

"QSO timeout — no response"

- The called station didn't hear you (too weak, bad timing, etc.)
- Try a different station, or switch to manual mode and check levels first
- See Troubleshooting for more

Next: Set up Time Sync for accurate FT8 timing, or explore the Web UI.

The Tab5 transmits FT8 via the QMX's `TA<freq>;` CAT command — no PC audio path, no WSJT-X. The QMX does all DDS synthesis and envelope shaping; the Tab5 sends the 79 tone-frequency commands at 160 ms cadence, bracketed by `TX;` / `TA0;` / `RX;`.

Replying to a station

1. In the decode list, **hold your finger on a row** for ~250 ms. A dim highlight appears after ~80 ms so you can see which row you're targeting. List scroll locks once the gate fires; drag up or down to land on the right row.
2. **Lift** — a confirmation modal shows the exact message that will go on air, the audio frequency, and the target slot parity.
3. Tap **Auto Pounce** to hand the full QSO to the auto-engine, or **Transmit** for a single manual message.
4. Tap **Cancel** (in the modal, or the armed indicator in the left pane) to disarm without transmitting.

Transmit is intelligent (v1.3.0): it builds the correct *next* message for that station from what they last sent — the same semantics as a WSJT-X double-click. Their CQ → your grid (or your report directly, if **Skip TX1** is on); their grid → your report; their report → `R`+your report; their `R`-report → `RR73`; their `RR73`/`73` → `73`. The report value is always your live measurement of their signal. You can walk an entire QSO step by step with nothing but Transmit taps — and when you send the closing `RR73`/`73`, the QSO **logs to ADIF** exactly like an auto-engine contact. Auto Pounce is offered on any first reply; mid-QSO rows get Transmit only.

Slot parity is set automatically — if you heard them on an EVEN slot, your reply goes on ODD so they're listening when you transmit.

The auto-engine works the full exchange: TX1 (grid) → wait for their report → TX2 (R+report) → wait for RR73/73 → TX3 (73) → DONE. At every step it re-sends the current message for up to 4 consecutive slots if the other station doesn't respond. If no reply comes after 4 slots, the QSO times out (orange status, tap to clear).

Skip TX1 (faster pounce). With **Skip TX1** enabled in the Filter editor, a pounce opens with your signal report immediately instead of the grid exchange — saving one round trip. If the station has already aged out of the decode list, it falls back to the normal grid TX1 automatically.

Working a pile-up

When you call CQ or work a run, more than one station may answer at once — and a caller who replies while you're mid-QSO with someone else used to vanish from the live decode list once they stopped transmitting. They're now held in a **Pileup** list so you don't lose them:

- Whenever callers are waiting, the **ADIF-log** button on the FT8 screen becomes a **Pileup** button in a distinct colour, reverting once the list empties.
- Tap **Pileup** to see everyone who has called you and isn't worked yet. Tap a station to work them (the same confirmation modal as a decode-list tap), or tap the ✕ to dismiss one.
- **Hold the button to open the ADIF log** — the log is always reachable this way, even while the button reads "Pileup" (v1.3.0; a one-time hint appears the first time the button flips).
- A station is removed from the list automatically once you start a QSO with them, and a just-completed contact's trailing 73 can't put them back.
- Worked-before stations appear in the pileup unless **Exclude worked-before** is checked — the pileup follows the same rule as the auto-answer, so dupes you're willing to work stay visible.

The tracker never transmits on its own — it only remembers callers; you choose who to work. If you'd rather it *did* transmit, check **Auto-work pileup** in the Filter editor: when your current QSO completes (or immediately, if you check it with callers already waiting and nothing else going on), the strongest waiting caller is pounced automatically, draining the pile one contact at a time. It carries the same unattended-TX warning as the robot.

Calling CQ — CQ-run mode

Tap **Call CQ**. No confirmation modal. The engine:

1. Scans the current decode list for occupied 50 Hz audio bins.
2. Picks the nearest unoccupied bin to 1500 Hz (standard FT8 sub-band centre).
3. Arms the CQ on the matching slot parity and waits for the boundary.

From there it's fully automatic:

- The opposite-parity slot is decoded while CQ runs so replies are never missed.
- The moment a station replies with your callsign, CQing stops and the exchange starts automatically — best-SNR caller chosen if multiple stations answer simultaneously.

- After the exchange completes (RR73 → 73 → logged), CQ resumes on the same frequency.
- A mid-exchange timeout also resumes CQ rather than dropping the frequency.
- While CQ-run is active, other stations' `CQ` rows are hidden so replies to you stand out.

Tap **Cancel** in the TX status bar at any time to stop.

Reply filter

Tap **Filter** in the left pane to open the filter editor. Controls which stations CQ-run auto-answers and which rows appear in the decode list.

- **Include 1 / Include 2** — if either field has text, only messages containing one of its terms are eligible. Space- or comma-separated (e.g. `POTA SOTA` or `JA, VK`), matched against the *whole* decoded message text so POTA/SOTA tags, grid squares, country prefixes, `/P` suffixes etc. all work.
- **Exclude 1 / Exclude 2** — messages containing any of these terms are skipped even if they'd otherwise match.
- **Exclude plain CQ callers** — hides bare `CQ ...` rows so only replies and exchanges remain visible.
- **Exclude worked-before** — hides callsigns already worked **on the current band** (per-band, from your ADIF log) from the list and from CQ-run auto-replies; the same station on a different band is treated as not worked.
- **Skip TX1** — when you pounce a station, open with your signal report instead of the grid exchange for a quicker QSO (falls back to the grid exchange if the station has aged out of the decode list).

Tap **Save** to persist (NVS) and apply immediately.

Auto-answer CQ (robot)

⚠ **Transmits unattended — never leave it running unsupervised.** When enabled, the robot picks a CQ caller every slot (filtered the same way as above, plus a worked-before skip) and runs the full exchange itself — no per-QSO confirmation, no tap required. Pick a **Priority**: Strongest signal, Weakest signal, or Most distant grid. Same TX1→report→RR73→73→ADIF-log flow as a manually-tapped reply; you just aren't the one tapping. You remain responsible for everything it transmits under your callsign, same as any other unattended digital-mode software.

ARRL Field Day mode

A checkbox in the same Filter modal as above (with Class/Section text fields next to it) switches the FT8 exchange from grid/signal-report to ARRL Field Day's class+section format — e.g. `16A EMA` instead of a grid square, using the standard `WA9XYZ KA1ABC R 16A EMA`-style FT8 message type (the same one WSJT-X uses for FD). Pounce and CQ-run both follow the convention automatically: the initial grid-exchange message is unchanged, but the report-equivalent step carries class+section instead, with the receiving side echoing it back `R`-prefixed.

While the mode is on, **Call CQ** automatically tags your CQ message `CQ FD <call> <grid>` (the same "CQ modifier" mechanism as `CQ POTA`/`CQ DX`) so other Field Day stations know to expect this exchange instead of a normal report — this overrides any other modifier on the active CQ preset for as long as the mode is enabled. The long-press CQ preset editor reflects this: while Field Day mode is on, the three presets are shown dimmed (their own modifier doesn't matter right now) and a live preview line shows exactly what will be transmitted.

Completed Field Day QSOs log the standard ADIF contest fields (see the table below) alongside the usual call/freq/time fields, so they import cleanly into contest-logging software.

FT8 Simulation mode

A **"FT8 Simulation Mode"** checkbox in the FT8 settings drawer (FT8 screen only) lets you practice everything — manual step-by-step Transmit, Auto Pounce, CQ-runs, pileups, and Field Day exchanges — with **no real station, no antenna, and no QMX connected at all** (v1.3.0; previously the radio had to be attached even though it was never keyed).

Six phantom stations (three US, three DX) call CQ on their own tones. Each phantom message is a real FT8 message, synthesized to actual GFSK audio and decoded through the same on-device receive pipeline real RF goes through — what lands in the decode list genuinely round-tripped the receiver. The phantoms behave like real operators:

- Tap a CQ to pounce (auto or fully manual — they answer either), or **Call CQ** yourself and **four of them answer at once**, building a genuine pileup to practice the pileup tools on.
- They're **patient**: each message repeats every cycle, up to four times, until you respond — then they give up and go back to CQing. A phantom you're mid-QSO with stops CQing; one you've worked stops answering your CQs for the session (toggle sim off/on to reset).

- Their replies match **what you actually transmitted** — grid gets a report back, a report gets a roger, `RR73` gets a courtesy `73`.
- The **Fast pounce** toggle (below) is honoured: with it ON, phantom messages surface just before the slot boundary; with it OFF, just after — so you can see exactly what the toggle changes.
- Swiping to the Panadapter and back clears the phantom rows and pileup for a fresh session.

While simulation mode is on, a **breathing red border** frames the whole screen as an unmissable reminder that nothing transmitted right now is real — the hard interlock lives in firmware (every CAT command that would key the radio is skipped, logged instead), not just in the UI. Completed simulated QSOs log to the same ADIF file as real ones — deliberately, so the logging/upload paths get exercised too. When you're done practicing, the ADIF viewer shows a **"Del N test"** button (only while simulation records exist — they're recognizable by their missing frequency): two taps wipes every practice contact from the log.

TX status indicator

The left pane shows a persistent status line below the slot countdown:

Colour	Meaning
Red	`TRANSMITTING: <message>` — burst in progress; tap to abort
Amber	`TX armed: <message> → EVEN/ODD, ~Ns` — waiting for slot; tap to cancel
Red-orange ⚠ FREQ BUSY	Armed tone is occupied (±50 Hz guard). During CQ-run this self-heals — the next cycle automatically hops to the nearest clear tone. Mid-exchange (replying to a specific station) the tone is locked to match them, so it just rides out until the exchange ends
Green	QSO complete
Orange	QSO timed out — tap to clear
Dim white	FT8 engine status passthrough (capturing / decoding / symbol count / ...)

QSO override buttons

During an active auto-QSO exchange (any state between first reply and final 73), three buttons appear in the left pane:

Button	What it does
Re-send / <field> (amber)	Re-arms the current outgoing message immediately. The label shows what will go out — e.g. **Re-send / JO45** when waiting for their report, **Re-send / -07** when waiting for RR73
RR73 (blue)	Skips straight to RR73, bypassing the report step
73 (green)	Fires the 73 sign-off and closes the QSO immediately

These are deliberate operator nudges for busy-band edge cases where the auto-engine falls a slot behind or you can see the exchange is further along than the state machine knows. The auto-engine handles the normal case; these exist for when a quick nudge is faster than waiting through the 4-slot retry cycle.

TX power / SWR readout

After each burst the Tab5 queries `PC;`/`SW;` for instantaneous forward power and SWR and shows the result briefly in the status line: `Last TX: X.XW SWRx.xx [Ns]`. If SWR > 4.0 (indicating the QMX SWR-protection latch tripped), the firmware automatically cycles `TX;` / 150 ms / `RX;` to clear the latch so the radio is ready for the next TX slot without operator intervention.

CQ message presets

Long-press the **Call CQ** button to open the preset editor. Three message slots with radio buttons — check the active one. A `+ <call> <grid>` quick-insert appends your identity. Standard CQ constructions (`CQ DX OZ1LAV JO65FR`, `CQ POTA …`) and ≤13-char free text both encode via the general ft8_lib encoder. Presets persist to NVS. The Call CQ button label shows the selected message; a short tap transmits it.

QSO logging (ADIF)

Every completed FT8 QSO — pounce or CQ-run — is automatically written to an ADIF v3.1.4 file at `/spiffs/qso.adi` on the Tab5's internal flash. The log survives reboots and re-flashes.

Download. The web UI bottom bar shows **"N QSOs ↓"** once the first QSO is logged. Clicking it downloads `qso.adi` for import into any logging software (WSJT-X, Log4OM, DXKeeper, ...) or for LoTW/POTA.app, which the Tab5 can't upload to directly (see below).

Upload to QRZ Logbook / eQSL — directly from the Tab5. Two more buttons appear next to the ADIF download link once you've logged a QSO: **"QRZ ↑"** and **"eQSL ↑"**. Tap either the first time and it prompts for credentials (QRZ: API key from *Logbook Data → API Key* on qrz.com; eQSL: your username and password — eQSL has no API-key scheme) and saves them on the Tab5, not just in the browser. Every tap after that uploads everything logged since the last successful upload — no need to re-enter credentials, and no risk of duplicate uploads, since the Tab5 tracks how far it's gotten into the log. If a record is rejected (bad credentials, quota, etc.) the run stops and reports the reason rather than skipping past it; fix the cause and tap again to resume from where it left off. eQSL accepts a whole batch of QSOs per request; QRZ's API takes one at a time, so a big log takes one HTTP round trip per QSO.

LoTW and POTA.app have no equivalent built-in path — LoTW requires a certificate-based TQSL signing step, and POTA.app has no upload API at all (browser login or email only). Use the ADIF download above for both.

Fields in each record:

ADIF field	Content
CALL	Their callsign
FREQ	Dial frequency in MHz (3 d.p., e.g. `14.074`)
BAND	Amateur band (e.g. `20M`)
MODE	`FT8`
SUBMODE	`FT8` (required by LoTW TQSL and eQSL for digital mode credit)
RST_SENT	Our SNR estimate (e.g. `-07`; `599` if unavailable)
RST_RCVD	Signal report received
QSO_DATE	UTC date `YYYYMMDD`
TIME_ON	UTC time `HHMMSS`
MY_CALL	Your callsign (from Identity in the drawer)
MY_GRIDSQUARE	Your grid
GRIDSQUARE	Their grid (from the decoded FT8 message)
CONTEST_ID, STX_STRING, SRX_STRING, ARRL_SECT, MY_ARRL_SECT	Field Day mode only: contest ID `ARRL-FD`, your/their literal `<class> <section>` exchange text, and their/your section alone

Clear. `GET /api/adif/clear` from the web UI wipes the file and resets the worked-call cache.

> Set your callsign and grid in the settings drawer → **Identity** before your first QSO so they appear correctly in logged records.

6. Time sync

FT8 requires accurate UTC time — within ±1 second of the real thing. The panadapter syncs time from multiple sources in priority order.

6.1. Time Sources (Priority Order)

1. **GPS-disciplined QMX** (auto-detected) — phase-locked to the GPS second, ~10 ms, works offline
2. **WiFi + SNTP** (if available) — ~10 ms, re-syncs every ~1 hour
3. **Tab5 RTC** (if set) — persists across power cycles (±1 min accuracy)
4. **FT8/FT4-derived** — offline fallback only (ignored while GPS/SNTP is up)
5. **Manual set** — enter time manually via the settings drawer

GPS and SNTP are the two accurate sources and are used whenever present; if you're offline with neither, the panadapter falls back to the RTC (or FT8-derived / manual).

6.2. Offline (POTA / Portable)

If you're operating **without WiFi** (POTA, portable, SOTA):

1. **Set the Tab5 RTC before you leave home** (settings → Time)
2. The RTC is powered by a **supercap battery** and holds time for **30–40 hours** without power
3. When you turn the Tab5 on in the field, it reads the RTC immediately
4. Optional: if your QMX has GPS, the panadapter syncs the RTC from QMX GPS every 5 minutes

No internet needed — FT8 timing works offline.

6.3. WiFi + SNTP

When WiFi is active:

1. The panadapter connects to an SNTP server (default pool.ntp.org)
2. Gets the current UTC time
3. Sets the system clock and writes it to the Tab5 RTC
4. Checks periodically (~hourly) and re-syncs if time has drifted

SNTP sync is **automatic** — you don't need to do anything. The bottom bar shows the current time (updates every second).

6.4. QMX GPS Time Sync (auto-detected)

If your QMX has **internal GPS** (QMX+ models often do), there is **nothing to enable** — the Tab5 detects it automatically:

1. At connect, it catches the QMX's `TM;` seconds *flip* (the true GPS second boundary) and compares it against SNTP.
2. If they agree tightly, the QMX is flagged GPS-disciplined and the clock is **phase-locked to that second edge** (~10 ms, drift-free), not just set to the whole second. Re-locks every 5 minutes.
3. If they don't agree (a non-GPS QMX with a free-running RTC), it's used only as a low-priority offline fallback.

The verdict is remembered across reboots. A GPS-disciplined QMX shows **UTC(GPS)** in the bottom bar and is a top-tier source (as good as SNTP); a non-GPS QMX shows **UTC(QMX)** and only helps when offline.

6.5. Manual Time Set

To set time manually:

1. Swipe ← to open the settings drawer
2. Tap **Time Sync** section
3. Tap **Set Manual Time**
4. Enter hours, minutes, seconds (UTC)
5. Tap **Apply**

The time is set immediately and written to the RTC.

6.6. FT8-Derived Sync (Offline Fallback)

The panadapter can also **estimate the UTC offset** from decoded FT8/FT4 signal timing and nudge the clock:

1. Decode several messages (requires on-air activity)
2. Measure their timing relative to the slot boundary
3. Estimate the error and apply a damped correction

This runs only as an offline fallback. While SNTP or a GPS-disciplined QMX is available they are authoritative and FT8-derived sync is **ignored** — deliberately. The FT8 slot offset the device measures is dominated by ~½ second of one-way *receive audio latency* (QMX SDR + USB buffering), which is **not** a clock error; letting it pull the clock would actually drag your transmit timing late. So it only touches the clock when you're off-grid with no SNTP/GPS.

Most useful when: WiFi is unavailable, no GPS QMX, and your RTC has drifted.

Fine-Tune Time with "Sync Time" (FT8 Only)

In **FT8 mode**, the Filter modal includes a **"Sync Time" button** that opens an interactive time-setting panel:

1. Tap the **Filter** button (FT8 screen, left pane)
2. Tap the **Sync Time** button (bottom right of the modal)
3. A panel appears with three fields: **[HH] : [MM] : [SS]**
    - **HH** / **MM** (hours/minutes) — tap to edit via numpad (0–23, 0–59)
    - **SS** (seconds) — auto-syncs from FT8 signals (blue frame = actively syncing, grey = locked)
4. Tap **Apply** to write the time to the RTC and system clock

The **SS field** updates automatically from decoded FT8 messages — each decode gives a sub-second correction estimate. Tap **SS** to toggle between auto-syncing (blue) and locked (grey). While locked, seconds still count at the captured offset, so you don't lose precision after locking.

Use case: You're operating portable without WiFi, your RTC is ~5 seconds off, and there's on-air FT8 activity. Set HH/MM manually, let SS auto-sync to the decoded signal timing for a few seconds, lock it, and you're done — FT8 timing is now precise.

FT8-derived sync shows **SS (sub-second)** in the bottom bar when active.

6.7. Bottom Bar Time Display

The center of the bottom bar shows the current UTC time and which source is active:

Indicator	Meaning	Updates
UTC(GPS)	GPS-disciplined QMX, auto-detected, phase-locked to the GPS second (~10 ms)	Every 5 min
UTC(NTP)	WiFi + SNTP	~1 hour
UTC(FT4) / **UTC(FT8)**	FT4/FT8-derived offline sync	Continuous (when decoding, offline only)
UTC(RTC)	Tab5 supercap RTC	At boot
UTC(MAN)	Manual time set	On-demand
UTC(QMX)	Non-GPS QMX RTC fallback (offline only)	Every 5 min
UTC	Fallback (no sync source)	—

The suffix tells you at a glance which source is **currently in charge** (the active authority, not merely the last one that wrote the clock). A GPS-disciplined QMX (`UTC(GPS)`) and WiFi/SNTP (`UTC(NTP)`) are both accurate — if you see either, you're set. If you're off-grid you'll see `UTC(FT8)`/`UTC(FT4)` (on-air digital activity), `UTC(QMX)` (non-GPS QMX clock), or `UTC(RTC)` (the supercap RTC you set before leaving home).

6.8. Slot Timing

FT8 operates on **15-second slot boundaries** aligned to UTC. The panadapter:

1. Reads the current system time
2. Waits until the next 15-second boundary (hh:mm:00, hh:mm:15, hh:mm:30, hh:mm:45)
3. Starts a 15-second capture window
4. Decodes the received FT8
5. Transmits (if armed) at the start of the next boundary

This alignment is **automatic** — you don't configure slots. But **time accuracy is critical**:

- ±500 ms error → can miss decodes or transmit off-slot
- ±1 s error → very few decodes, transmit often off-time

- ±2 s error or worse → FT8 doesn't work

6.9. Time Sources Summary

Source	Accuracy	Updates	Works Offline?
QMX GPS	~10 ms (auto-detected, tick phase-lock)	every 5 min	Yes (if QMX has GPS)
SNTP	±10 ms	~1 hour	No
RTC	±1 min	at boot	Yes (30–40 h)
Manual	±1 sec	on-demand	Yes (until power cycle)
FT8-derived	offline fallback only	continuous	Yes

Next: Configure Settings or troubleshoot Time Sync Problems.

The Tab5 needs accurate UTC for FT8 slot timing. Sources in priority order (highest first):

Source	When applied
SNTP (WiFi)	Always authoritative whenever WiFi is connected and has synced at least once — sets system clock, Tab5 RTC, and NVS
Tab5 RTC	RX8130CE supercap-backed (~30–40 h retention) — applied at boot before QMX or WiFi are available
QMX `TM;`	Offline fallback only — applied when WiFi is down or has never synced (field/POTA with no WiFi)
FT8 signal timing	Sub-second correction from the decoded signal population's average timing, auto-applied continuously every slot while FT8 is decoding (damped, so a noisy single slot can't yank the clock) — deliberately tracks who you're actually trying to work, not absolute GPS/NTP truth. A manual one-shot version is also available via the **Sync Time** modal (see below)
Manual	Full HH:MM override in the **Sync Time** modal — for rare POTA sessions with neither WiFi nor QMX clock

Each accepted sync writes through to the RX8130CE so the clock persists across power-off. The Tab5 also polls `TM;` every 5 minutes in the background to catch QMX GPS lock events.

For POTA / no-WiFi use: uncheck **WiFi initiated** in the settings drawer. Time comes from the QMX on USB connect (or the supercap RTC if the Tab5 was recently synced). FT8 slot timing will be accurate from either source.

FT8 time calibration modal

On the FT8 screen, tap **Filter** → **Sync Time** to open the time calibration modal. It shows three large boxes: **[HH] : [MM] : [SS]**.

- **HH / MM** — pre-filled from the current UTC clock. Tap either box to edit it with a numpad. Use this for rare POTA situations where neither WiFi nor the QMX clock is available.
- **SS** — auto-syncs continuously from decoded FT8 signals. Each slot, every successfully decoded station contributes a timing sample; a robust outlier-rejecting average (median ± a tolerance window) across all of them sets the correction, so one bad decode can't throw off the reading. The box border flashes bright blue for ~30 ms each time a fresh measurement lands — a visual heartbeat that sync is active — and the hint below reads **"Flash: FT8 synced..."**. A blue frame means it is actively tracking; tap SS to lock it.
- **Apply** — if only SS was synced (HH and MM untouched), `time_sync_apply_correction_ms()` nudges the system clock by the measured sub-second offset and writes through to the RTC. If HH or MM was edited, `time_sync_set_manual()` sets the full time. The bottom bar shows **UTC(FT8)** after an FT8-derived correction, or **UTC(manual)** after a full manual set.

---

7. Reference

Gestures

Gesture	Where	Effect
Tap	Spectrum / waterfall	Tune to tapped frequency (snapped)
Touch + drag (hold ~250 ms first)	Spectrum / waterfall	Cyan cursor snaps grid-to-grid; tunes on release
Fast one-finger horizontal swipe	Spectrum / waterfall, any zoom	Pan/"stroll" the view; retunes to the new centre on release
Double-tap	Spectrum / waterfall	Reset zoom and pan to ×1.0 / centred
Pinch (two fingers)	Spectrum / waterfall	Zoom ×1.0–×24.0
Two-finger drag	Spectrum / waterfall (zoomed)	Pan the zoomed window
Swipe → from left edge	Left edge strip	Toggle Panadapter ↔ FT8 screen
Swipe ← from right edge	Right edge strip	Open settings drawer
Tap right grip handle	Right edge	Open settings drawer (alternative)
Swipe →	Open drawer, or spectrum while drawer open	Close settings drawer
Swipe ↑ from bottom edge	Bottom edge strip	Open memory channel picker
Tap or swipe ↓	Top-bar item (Band/Mode/BW/Freq/Zoom)	Open that item's selector

Gesture	Where	Effect
Touch and hold ~250 ms	FT8 decode list row	Dim preview at ~80 ms; full selection at 250 ms (scroll locks, drag moves highlight, lift to confirm)
Quick swipe	FT8 decode list	Scroll the list normally

Per-unit IF calibration

The QMX's +12 kHz IF injection varies slightly between units. If signals appear consistently shifted left or right of where the QMX is actually tuned, open the settings drawer → **IF calibration** slider (±100 Hz, persisted to NVS). Default 0. Reported by Ken KF0AYY, whose unit needed about −55 Hz to centre correctly.

Hardware

- **M5Stack Tab5** — ESP32-P4 v1.3 (ECO2), ST7121 or ST7123 5" 720×1280 MIPI-DSI touch display, 32 MB PSRAM, ESP32-C6 co-processor for WiFi
- **QRP Labs QMX or QMX+** — firmware 1.03.002 or newer (the 1.04.002 beta also works, and unlocks AM mode + Antenna Tune)
- **USB-A → USB-C data cable** between Tab5 USB-A host port and QMX (full data, not charge-only)
- **USB-C power supply** for the Tab5 (5 V / 2 A or better, or internal battery)

```
┌──────────────┐  USB-A → USB-C (data cable)    ┌────────────┐
│  M5Stack     │ ──────────────────────────────►│   QMX+     │
│  Tab5        │  UAC: I/Q audio (48 kHz stereo)│            │
│              │  CDC-ACM: CAT control          └────────────┘
│              │
│              │  USB-C (any power cable)       ┌────────────┐
│              │ ──────────────────────────────►│ 5 V source │
└──────────────┘                                └────────────┘
```

What works without the QMX connected

Works without QMX	Needs QMX connected
UI, settings drawer, gestures	Spectrum + waterfall (no I/Q = flat line)
WiFi, web UI, screenshots	Top bar Band / Mode / BW
Callsign / grid / WiFi credentials	Tap-to-tune, mode popup (CAT writes)
Brightness, colour map, dB range	S-meter, FT8 decode/TX, QMX firmware readout

If the spectrum is flat and the top bar shows `---`, that is almost always the radio not being seen over USB — which loops back to the cable (or the QMX being off / in flash mode).

Spectrum signal shifted / mirrored, tunable across the whole 48 kHz window

This means the QMX never confirmed IQ mode for the session — without it the radio streams plain (non-IQ) audio instead of a centred baseband, so the signal appears at the wrong place and slides across the full 48 kHz window as you tune. As of v0.19.3 the Tab5 retries the IQ-mode handshake automatically at connect (up to 4 attempts) and this resolves itself almost every time; if it still fails after all retries, a red banner appears across the top of the screen telling you so immediately, instead of leaving you to figure it out from a shifted waterfall. If you see the banner, power-cycle the QMX (forces a fresh handshake on reconnect) or check the QMX's own **System Config → IQ Mode** setting.

Reporting hardware issues

Near the top of the boot log you will see:

```
I (xxxx) bsp_info: === TAB5 BSP INFO ===
I (xxxx) bsp_info: chip:     ESP32-P4 rev v1.3
I (xxxx) bsp_info: psram:    30 MB
I (xxxx) bsp_info: panel:    ST7123 (inferred from touch)
I (xxxx) bsp_info: touch:    ST7123 @ 0x55
I (xxxx) bsp_info: heap:     230.5 kB internal free, 28.80 MB PSRAM free
I (xxxx) bsp_info: idf:      v5.4.4
I (xxxx) bsp_info: firmware: v0.21.0
I (xxxx) bsp_info: =====================
```

When opening a hardware issue, paste this block — the `panel` and `touch` lines are the first thing needed to identify which Tab5 hardware revision you have.

> **Note on `reset_reason`:** a deliberate reset-button force-off returns `panic/exception`, not `external-pin/power-on`. A genuine firmware crash always prints a `Guru Meditation` register + backtrace dump immediately *before* the reboot. No backtrace = abrupt power-off, not a crash.

Appendix — Technical Reference

The sections below are the remaining, more technical parts of the same project README (build instructions, internal design notes, roadmap) — included here so the links above resolve, rather than dead-ending. Skip ahead if you only want operating instructions.

Appendix A — Build from source

Requires **ESP-IDF v5.4.4** — pinned; do not upgrade. ESP32-P4 v1.3 silicon requires `CONFIG_ESP32P4_REV_MIN_0=y` and CPU capped at 360 MHz, both baked into `sdkconfig`.

```
# Activate the IDF environment, then:
idf.py build flash monitor
```

Or with the `qmx` PowerShell helper — add to your `$PROFILE` and adjust the COM port:

```
function qmx {
    param([string]$cmd = "fm")
    if (-not $env:IDF_PATH) { & C:\esp\v5.4.4\esp-idf\export.ps1 }
    switch ($cmd) {
        "b"   { idf.py build }
        "f"   { idf.py flash }
        "m"   { python -m esp_idf_monitor -p COM3 build\qmx_panadapter.elf }
        "fm"  { idf.py flash; python -m esp_idf_monitor -p COM3
         build\qmx_panadapter.elf }
        "bfm" { idf.py build flash; python -m esp_idf_monitor -p COM3
         build\qmx_panadapter.elf }
    }
}
```

Exit monitor: `Ctrl+T` then `Ctrl+X`.

Appendix B — Under the hood

I/Q balance correction

The QMX presents I/Q audio with a small but measurable amplitude imbalance and quadrature error between channels. Without correction, every signal produces a mirror image across the centre frequency (~30 dB weaker), cluttering the waterfall on a busy band.

A blind adaptive Gram-Schmidt orthogonaliser runs sample-by-sample in the audio pipeline before the FFT. It maintains running estimates of:

- **DC offset** on each channel (τ = 1 s)
- **Power** in I and Q (τ = 200 ms)
- **Cross-product** I·Q (τ = 1 s) — non-zero cross-product means the channels are not orthogonal

Correction per sample:

```
i_out = i
q_out = (q − K_phi · i) × K_amp
```

No calibration step needed; the estimator converges on ambient band noise within a few seconds of any signal. A two-speed startup runs all alphas at 8× for the first 2 s of real signal after each reset (effective convergence ~125 ms), then drops to steady-state. Toggle in the settings drawer; re-enabling resets the estimator so it reconverges from a clean state.

DSP pipeline

- **FFT:** 1024-pt complex, Blackman-Harris window. esp-dsp ANSI fallback is used (the PIE/vector version crashes under sustained WebSocket load on this silicon).
- **IF offset:** The QMX presents I/Q at +12 kHz; spectrum, waterfall, and S-meter all shift bin selection by `n_bins/4` so the VFO signal appears at the visual centre.
- **DC blocker:** one-pole IIR on the I/Q stream before FFT.
- **Spectrum smoothing:** per-bin EMA (α = 0.4 default, adjustable).

- **dBm calibration:** `DSP_DB_CALIBRATION_OFFSET = -148.0 dB` measured on dummy load; noise floor reads −130 dBm; S9 = −73 dBm.
- **Waterfall scroll:** 1280×824 double-height canvas (~130 µs/tick vs ~92 ms with memmove).
- **Audio task:** polling on core 0 with a drain loop. Event-driven reads caused noise-floor pumping (~13 s cycle) from truncated UAC chunks; this architecture eliminates it.

Quirks and trade-offs

PI4IO expander must be initialised before display bring-up. The Tab5's PI4IO I/O expander holds `LCD_RST` and `TP_RST` low at chip power-on. Without explicit init, on a true cold boot the panel never comes out of reset and the DSI FIFO hangs. Soft resets mask this because the expander retains state across ESP32 resets. `display_init` calls `bsp_i2c_init()` + `bsp_io_expander_pi4ioe_init()` first, then waits 120 ms.

Patched components in `components/`. `espressif__usb_host_uac/` has `create_background_task = true` — required for UAC and CDC-ACM to coexist on the same USB host. `espressif__esp_lcd_touch_st7123/` makes `FW_VERSION_REG` the only mandatory register read (ST7121 NACKs the others and also adds a `max_touches > 10` bounds clamp). Do not replace either with the registry version without re-applying the patches.

LVGL software rotation (~50% FPS cost). The panel is natively portrait; landscape is achieved via `lv_display_set_rotation(LV_DISPLAY_ROTATION_90)`. Every flush goes through `rotate90_rgb565`. ~13 fps landscape vs ~22 fps portrait — acceptable for a panadapter. PPA hardware rotation conflicts with the USB host stack over DW-GDMA channels and silently kills UAC + CDC-ACM; do not set `CONFIG_LVGL_PORT_ENABLE_PPA=y`. A full native-portrait rewrite would recover the lost FPS but isn't planned — accepted as a permanent trade-off.

IDLE watchdog disabled. The LVGL rotation pipeline keeps CPU0 busy past the default watchdog window. `CONFIG_ESP_TASK_WDT_CHECK_IDLE_TASK_CPU0/CPU1` are off; the app-task watchdog (30 s) is still active.

Cross-thread CAT writes must go through the poll task. Writing CAT commands directly from the LVGL thread races the FA/MD/FW poll on the same CDC pipe — commands interleave and the QMX returns `?;`. Pattern: stash the request in a `volatile` and let `poll_task` drain it on its next cycle. See `s_pending_ssb_bw` / `cat_request_ssb_bandwidth()` in `cat.c`.

SSB filter bandwidth needs three coordinated writes. `MMSSB|Filter RX=<hz>;` commits the value (persists, shows in QMX menu); `MMSSB|Bandwidth=<hz>;` applies it live; `FW;` polling must be suspended while the width is pinned because reading the filter makes the QMX revert it. Dead ends: `FW<nnnn>;` CAT set returns `?;`; re-asserting the same mode digit does not reload the filter; `Bandwidth` write alone reverts on the next poll.

HTTPS needs mbedtls allocating from PSRAM, not internal RAM. The QRZ/eQSL upload features (v0.16.2) are this firmware's first-ever outbound HTTPS connections — SNTP, the only prior network client, is UDP. With the default `CONFIG_MBEDTLS_MEM_ALLOC_MODE=MBEDTLS_INTERNAL_MEM_ALLOC`, the TLS handshake's 16 KB+ buffers competed for the ~200 KB internal DRAM already under pressure from USB host/audio/FFT/LVGL and every connection failed with `ESP_ERR_HTTP_CONNECT`. Fixed by switching to `MBEDTLS_EXTERNAL_MEM_ALLOC` in `sdkconfig`, which moves those buffers into the 28 MB of free PSRAM instead.

Project layout

```
main/
  main.c                     app_main, task launch, orchestration
  display/display.c          BSP bring-up (PI4IO + LCD + touch)
  ui/ui.c                    LVGL widgets, touch handler, canvases
  ui/ft8_screen_view.c       FT8 decode list, touch-drag row selection
  ui/ft8_tx_modal.c          TX confirmation modal
  cat/cat.c                  USB CDC-ACM + Kenwood CAT
  audio/audio.c              USB UAC + ring buffer producer
  dsp/dsp.c                  FFT, spectrum mutex, DC blocker
  dsp/iq_balance.c           Gram-Schmidt I/Q correction
  ft8_tx.c                   FT8 TX engine (build/arm/run/abort)
  ft8_test.c                 FT8 slot loop (RX decode / TX burst)
  ft8_qso.c                  Auto QSO state machine
  ft8_sim.c                  FT8 simulation mode: phantom-station practice QSOs
  ft8_status.c               Mutex-protected FT8 status string
  adif/adif_log.c            ADIF QSO logging
  adif/qrz_upload.c          QRZ Logbook API upload (one record per request)
  adif/eqsl_upload.c         eQSL.cc upload (batched, multiple records per request)
  rtc/rtc.c                  RX8130CE supercap RTC driver
  time_sync/time_sync.c      Time sync orchestrator
  render/render.c            30 Hz render task, EMA smoothing
  render/render_waterfall.c Waterfall scroll (double-height canvas)
  screenshot/screenshot.c    RGB565 framebuffer capture for /ss.bmp
  storage/settings.c         NVS-backed settings with debounced flush
  wifi/wifi.c                C6 co-processor WiFi + SNTP
  net/webserver.c            HTTP server + WebSocket
  util/fps.c                 FPS counter
  util/diag_log.c            Diagnostic log ring buffer (vprintf hook)
```

Appendix C — Roadmap

The full per-version changelog — every release from v0.1.0 onward — lives in **docs/version-history.md**. This section tracks only what's planned next.

Next up

v1.3.0 is here — smarter manual FT8 operation from Roy KI0ER's field feedback: an **intelligent Transmit button** (sends the correct next message, WSJT-X double-click style), faster reply timing, a **cold-pounce early-decode toggle**, the pileup/ADIF-log lockout fixed, a fully rebuilt **practice simulator** that needs no radio at all, **USB mouse** support (radio unplugged), and a **web file browser** for the microSD card. Next on the bench:

- **Web-UI audio streaming.** Listen to the receiver in any browser on your LAN — demodulated on the Tab5, no PC. Already working in development; held back for quality tuning and an overnight streaming soak. Server mode (screen off, device just serves) rides along.
- **CW page.** Canned-message CW TX memories first; decoded-CW display after (the QMX decodes internally — mirroring it over CAT looks cheap).
- **User Manual polish.** Inline images (phase 2 — currently text-only), and a longer soak of the microSD "Save offline" path on the fragile WiFi link.

Longer term

- **CW decoder.** Goertzel-based, text scrolling under the spectrum. The QMX already does this internally — question is whether to mirror its output via CAT or run a parallel decoder on the Tab5.
- **Tab5 speaker / headphone audio.** Demodulated CW/SSB passband audio out of the Tab5's own jack, so the operator can monitor without the QMX's audio path.
- **Extended waterfall history.** PSRAM has room for several minutes of scrollback; two-finger drag to scrub through history.
- **QMX (small) support.** Same UI, different USB endpoint config and band table.
- **JS8 / RTTY modes.** See `docs/js8-feasibility.md` and `docs/rtty-feasibility.md`.
- **DSP polish.** Noise reduction, auto-notch.

Appendix D — Related projects

- DX-FT8 by Barb (WB2CBA) — open-hardware FT8 tablet transceiver; an inspiring reference for a similar use-case
- `qrp_companion` by Zhenxing Han (N6HAN) — Tab5 companion for QMX with audio + CAT; source of the polling audio task pattern and battery readout approach
- `ft8_lib` by Karlis Goba — FT8 encoder/decoder vendored as `components/ft8_lib`

Glossary

Common terms and acronyms used in this guide:

Term	Meaning
CAT	Computer-Aided Transceiver — radio control protocol (Kenwood-style commands via serial/USB)
CDC-ACM	Communications Device Class / Abstract Control Model — USB standard for serial ports
CQ	General call to any station (not directed at anyone specific)
CW	Continuous Wave — Morse code mode
DSP	Digital Signal Processing — mathematical signal analysis and filtering
FFT	Fast Fourier Transform — algorithm to convert time-domain audio into frequency spectrum
FT8 / FT4	Digital modes for weak-signal HF communication (15-second vs 7.5-second slots). Both fully supported (FT4 re-enabled in v0.21.0).
GPIO	General-Purpose Input/Output — microcontroller pins for digital signals
I2C / SPI	Serial communication protocols for connecting peripherals (sensors, displays, etc.)
IQ	In-phase / Quadrature — stereo representation of RF signals (real + imaginary parts)
LVGL	Light and Versatile Graphics Library — open-source embedded UI toolkit used for the display
NVS	Non-Volatile Storage — persistent memory on the ESP32 (survives power cycles)
PSRAM	Pseudo-SRAM — extra RAM on the Tab5 (used for large buffers like waterfall history)
QMX / QMX+	QRP Labs HF transceiver — the radio this panadapter controls and receives audio from
QSO	Radio contact / conversation between two stations

Term	Meaning
RTC	Real-Time Clock — battery-backed timer on the Tab5 (keeps time during power-off)
SNTP	Simple Network Time Protocol — synchronizes system clock via WiFi/internet
SWR	Standing Wave Ratio — antenna impedance matching metric (1.0 = perfect)
TX / RX	Transmit / Receive — keying the radio and listening
UAC	USB Audio Class — standard for streaming audio over USB
USB	Universal Serial Bus — physical connector and protocol (carries both audio and CAT commands)
UTC	Coordinated Universal Time — timezone-independent time standard for FT8 slot alignment
VFO	Variable Frequency Oscillator — the radio's tuning dial / frequency setting

---

Contributing

This is a solo project — all coding is done by the author (OZ1LAV) at his own pace, and **pull requests are not being accepted right now**. Bug reports, feature requests, and field reports are very welcome, though, via GitHub Issues or the QRPLabs Groups.io thread. See CONTRIBUTING.md for details.

Appendix E — License